

HOME BURGUER

Business Process Management Report



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Introduction

This report examines the Order-to-Cash (O2C) process at Home Burger's Lisbon Center store, with the goal of identifying inefficiencies and recommending improvements. As the food delivery market becomes increasingly competitive, Home Burger faces the challenge of optimizing its internal processes to enhance operational performance and customer experience. This is why the business aims to cut down on delays, get rid of mistakes, and adjust to the rising demand for delivery services.

The focus of this report is the O2C process, which includes order placement, payment, and delivery. This report analyzes the current workflow of the process (AS-IS), identifies problems, and proposes an improved version (TO-BE). Both quantitative and qualitative methods are used in the analysis, which is backed by simulation tools and business process modeling with BPMN. The redesign aims to increase productivity, adaptability, and service quality, while keeping expenses reasonable.

By implementing these suggestions, Home Burger will be able to manage the growing demand better and keep a competitive advantage in the evolving food delivery industry.

Background

Founded in 2016, Home Burger has grown into a well-known burger delivery company with six stores across Portugal. Initially starting in Lisbon, the business expanded to Porto, Santarém, and Portimão over the years. With over 100 employees and a steady customer base, Home Burger manages thousands of orders monthly through multiple channels such as phone, website, and in-store visits.

Despite its success, the company operates in an increasingly competitive market. The surge in delivery services after COVID-19 and evolving customer expectations have created pressure to improve operational efficiency. Each store currently manages its orders and customer data independently, often leading to delays, errors, and resource constraints.

This project focuses on the Lisbon Center store, which handles the highest volume of orders. By analyzing the current Order-to-Cash (O2C) process, we aim to identify inefficiencies and explore how process redesign can enhance Home Burger's ability to deliver excellent service while staying competitive.

Initial Assumptions

In the initial phase, several assumptions were necessary to facilitate a detailed analysis and process modeling due to undefined parameters in the project description.

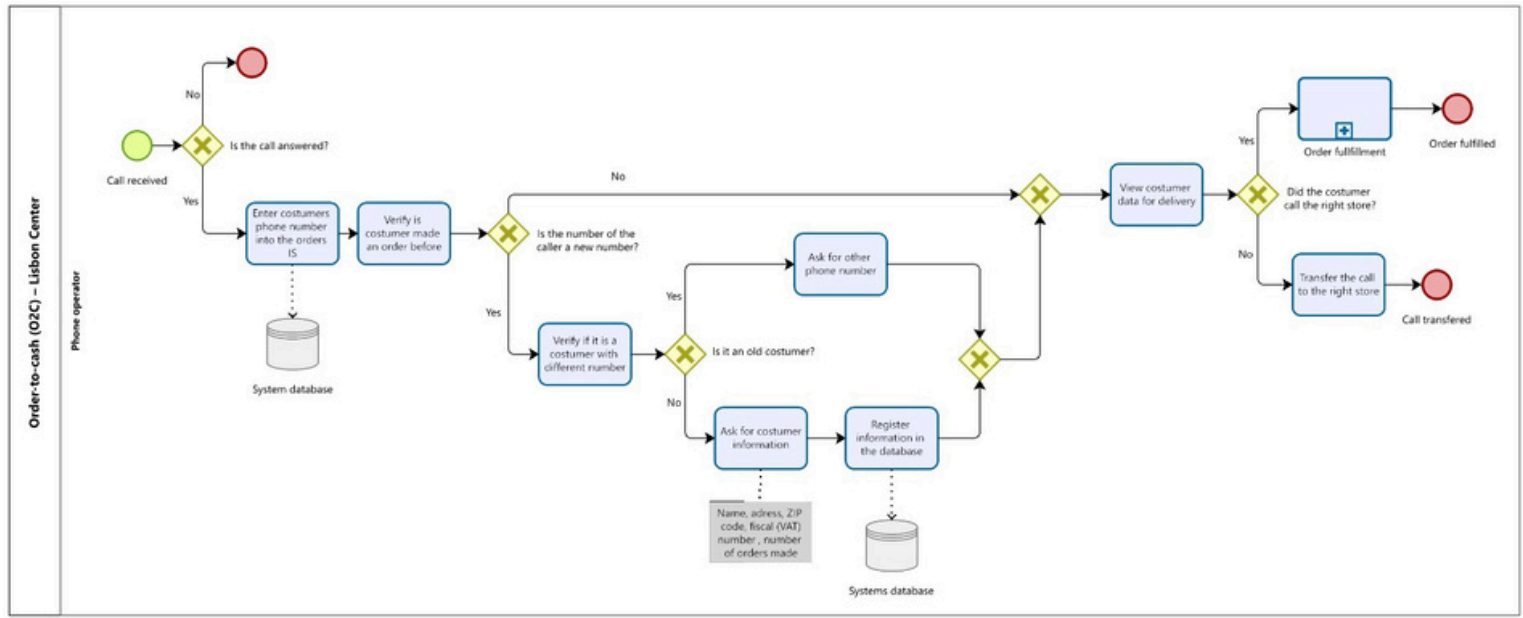
Firstly, we assumed specific salary figures for roles essential to the process, such as the manager, as these were not provided in the project documentation. For example, salaries for various roles, including phone operators, kitchen staff, couriers, and reception employees, were derived from general cost information given.

Secondly, where exact process times were not specified, we established average completion times based on either analogous tasks in the process or reasonable estimates drawn from industry benchmarks. For example, the average time for customer interactions during phone and in-person orders was standardized based on the provided task durations for related processes. This step was taken to ensure the consistency required to simulate and evaluate the current (AS-IS) and redesigned (TO-BE) processes.

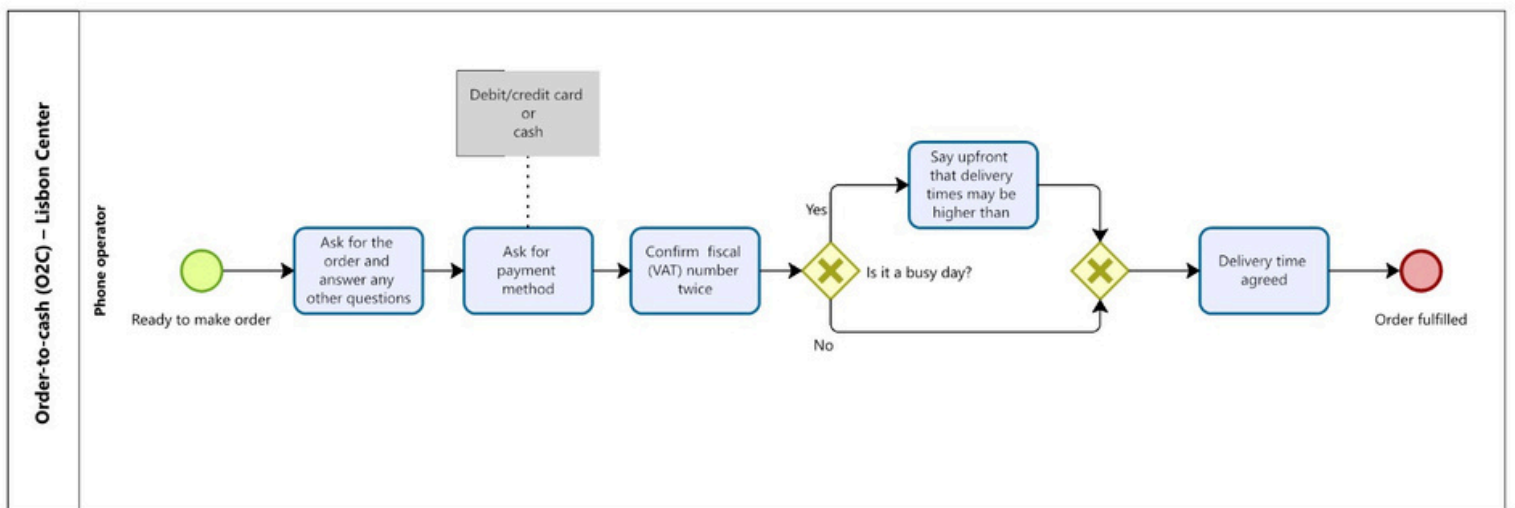
Lastly, we assumed that all tasks follow a normal truncated distribution to reflect realistic variability while avoiding extreme outliers. Additionally, start events, such as incoming customer orders, were modeled using a Poisson distribution to capture the random and time-dependent nature of these occurrences.

AS-IS Process Model

Orders registered by phone



Order Fulfillment Sub-process



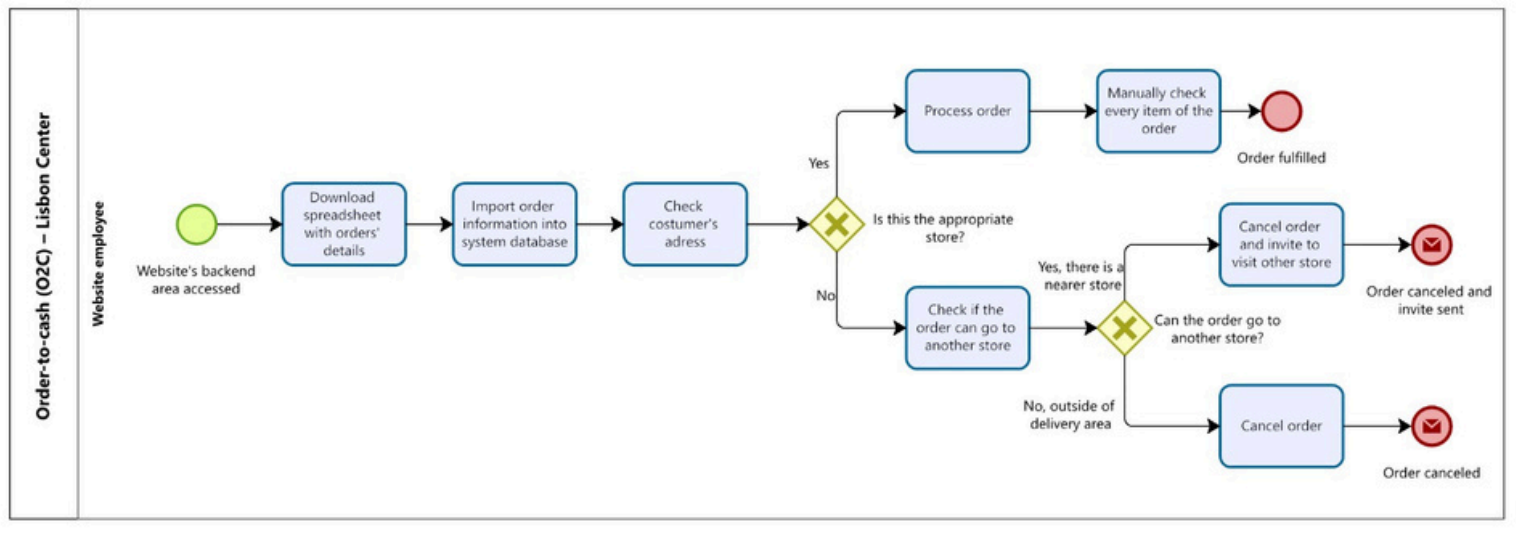
The process begins when a customer places a call to the store. The system checks if the call is answered. If not, the process ends. For answered calls, the operator inputs the customer's phone number into the system database. The system then checks if the customer has previously placed an order. If the number is recognized but associated with a different account, the operator verifies if it belongs to the same customer under a different number.

For new or unrecognized numbers, the operator gathers details such as the customer's name, address, ZIP code, and VAT number, registering them in the system database. The process then checks if the call is directed to the correct store. If not, the call is transferred to the appropriate location. After ensuring the correct store is handling the call, the order is processed, and the details are stored in the system. This step transitions the process to the "Order Fulfillment Sub-process."

In the Order Fulfillment Sub-process, the operator confirms the order details with the customer, addressing any additional questions. The customer then selects their preferred payment method—cash or card. The VAT number is verified twice for accuracy, as per the company's protocol.

The operator checks if it is a busy day, such as during peak hours or special events. If so, the customer is informed about potential delivery delays. Once delivery times are agreed upon, the order is finalized and marked as ready for fulfillment. This concludes the phone-based order process, ensuring seamless communication and preparation for subsequent delivery stages.

Orders registered by the website

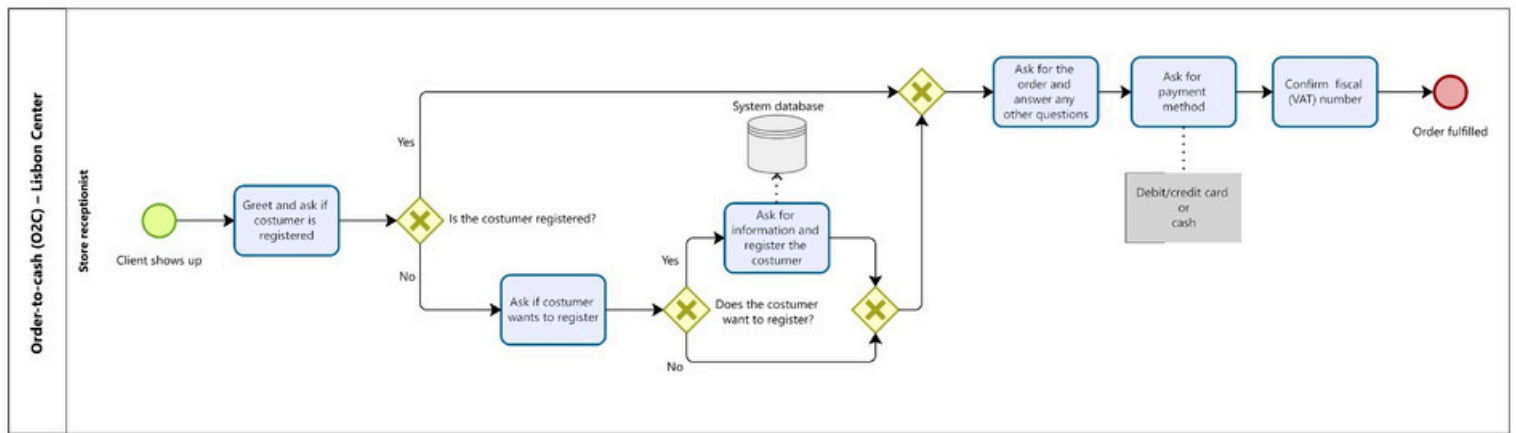


The process begins when a website employee accesses the backend area to retrieve a spreadsheet with the order details. These details are imported into the system database, ensuring that all information is ready for further processing.

Next, the employee verifies the customer's address to determine if the order can be processed by the store. If the store is appropriate for the customer's address, the order proceeds to be processed. During this step, the employee manually checks every item in the order for accuracy before finalizing the order, marking it as fulfilled.

If the address is not appropriate for the store, the system checks if the order can be directed to another store. If there is a nearer store that can handle the order, the order is canceled at the current store, and the customer is invited to visit or place an order with the appropriate store. If no nearby store can fulfill the order because it falls outside the delivery area, the order is canceled outright. This concludes the website-based order registration process.

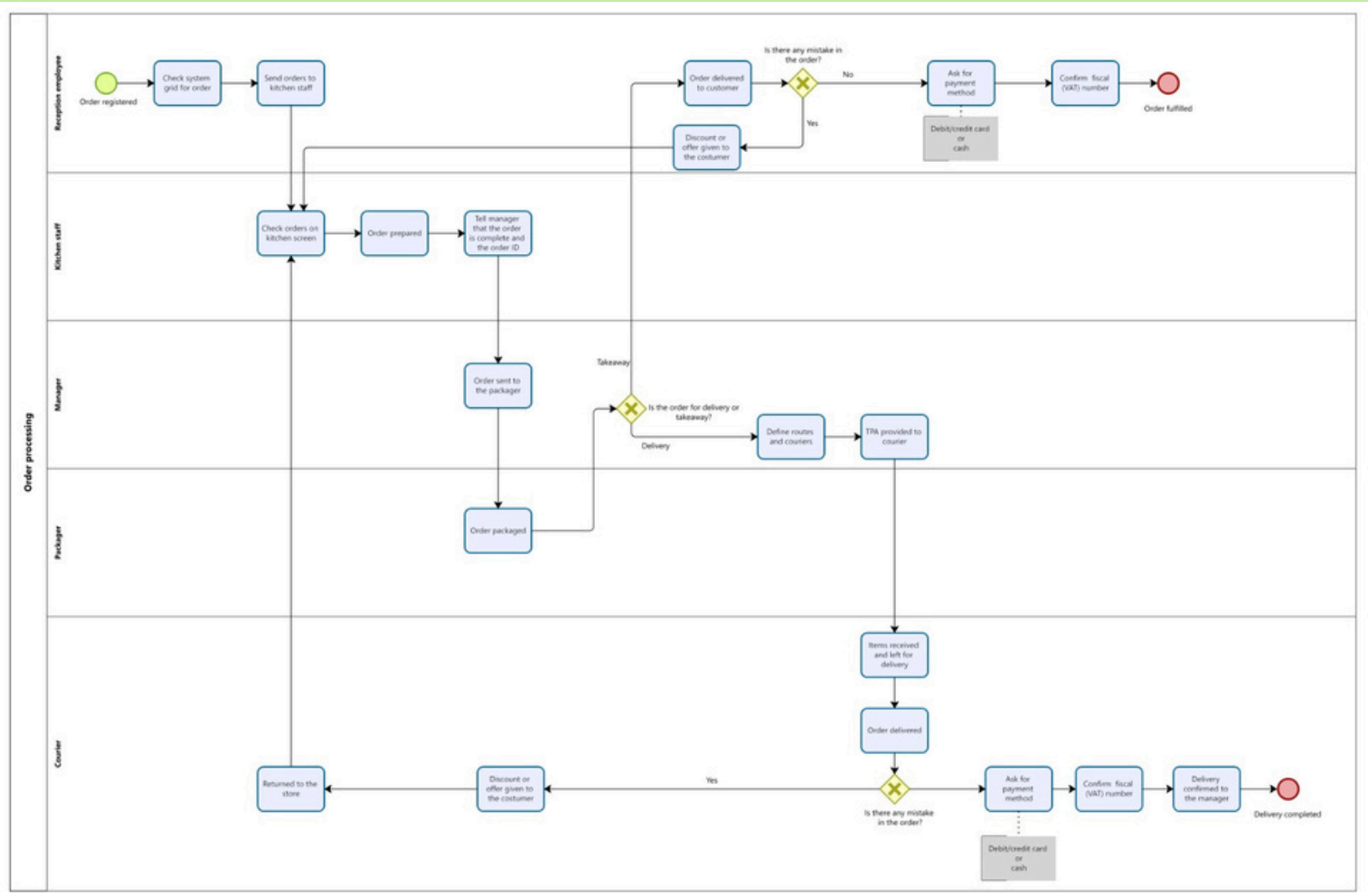
Orders registered in person at the store



The process begins when a client arrives at the store. The store receptionist greets the customer and asks if they are already registered in the system. If the customer is registered, the process continues directly to the order-taking phase. If the customer is not registered, they are asked if they would like to create a profile. For those who agree, their information, including name, address, and VAT number, is collected and entered into the system. If the customer does not wish to register, the process skips this step and proceeds to order-taking.

After registration (or skipping it), the receptionist asks the customer for their order and answers any additional questions. The payment method is then chosen, either cash or card, followed by verifying the fiscal (VAT) number twice to ensure accuracy. Once all details are confirmed, the order is finalized and marked as fulfilled. This completes the in-person order registration process.

Order processing



The process starts with the kitchen staff checking the system grid for orders, preparing the order, and notifying the manager upon completion. The order is then sent to the packager for proper packaging.

- Takeaway Orders: The packaged order is handed to the customer.
- Delivery Orders: Routes and couriers are assigned by the manager, and the courier receives the order and TPA for payment processing.

During delivery, any mistakes (e.g., missing items) result in the courier returning to the store for corrections, with potential discounts offered to the customer. If no issues occur, the courier delivers the order, collects payment, confirms the VAT number, and informs the manager upon completion. The order is then marked as fulfilled.

Qualitative Analysis

Value-Added Analysis

To improve the Order-to-Cash (O2C) process at Home Burger, it is crucial to evaluate each task based on its contribution to value creation. The Value-Added (VA) Analysis categorizes activities into three groups:

- **Value-Adding (VA):** These are activities that directly contribute to the customer's satisfaction, enhancing the final product or service.
- **Business Value-Adding (BVA):** Necessary activities that support business operations but do not directly add value from the customer's perspective.
- **Non-Value-Adding (NVA):** Tasks that do not contribute to value and create opportunities for process improvement or elimination.

This analysis will help identify areas where Non-Value-Adding tasks can be minimized or eliminated, allowing Home Burger to focus on enhancing Value-Adding activities while improving operational efficiency and customer experience.

Orders registered by phone analysis

TASK	TYPE	JUSTIFICATION
Enter customer phone number into system	VA	Necessary to retrieve or create customer information.
Verify if the customer made an order before	VA	Helps streamline the process for returning customers.
Is the number new?	BVA	Necessary to differentiate customer but does not directly add value.
Verify if it is a customer with another number	BVA	Important for maintaining accurate records but not value-adding to the customer.
Ask for customer information	VA	Critical for processing the order and ensuring delivery accuracy.
Register information in the database	VA	Ensured customer details are stored for future transactions.
Confirm delivery data	VA	Ensured order accuracy and timely delivery.
Did the customer call the right store?	BVA	Needed to direct orders correctly but adds no direct customer value.
Transfer the call to the correct store	NVA	Repetitive task that wastes time and resources; could be eliminated with automation.

Orders registered by the website analysis

TASK	TYPE	JUSTIFICATION
Access website backend	VA	Essential to receive customer-submitted orders.
Download spreadsheet with orders' details	BVA	Necessary but adds no direct value for customers; automation could optimize it.
Import order information into the system	VA	Critical for preparing orders for processing.
Check customer address	VA	Ensured orders are deliverable within the correct service area.
Is this the appropriate store?	BVA	Important for routing but could be automated.
Check if the order can go to another store	BVA	Reduces waste for the business but adds no value for customers.
Cancel order and invite to visit store	BVA	Operationally necessary but results from inefficiencies in the earlier steps.
Process the order	VA	Prepared the order for fulfillment.
Manually check every item in the order	NVA	Redundant task that could be automated to save time and reduce errors.

Orders registered in person at the store analysis

TASK	TYPE	JUSTIFICATION
Greet customer	VA	Improves the customer experience and builds trust.
Check if customer is registered	VA	Necessary for identifying returning customers.
Ask if the customer wants to register	BVA	Adds value to the business but does not directly benefit the customer.
Ask for information and register the customer	VA	Ensured accurate data for processing orders and future transactions.
Ask for the order and answer questions	VA	Addresses immediate customer needs and concerns.
Ask payment method	VA	Essential for completing the transaction.
Confirm VAT number	BVA	Ensures compliance but offers no immediate value to the customer.

Orders processing analysis

TASK	TYPE	JUSTIFICATION
Check system grid for order	VA	Ensures accurate task assignment for the kitchen staff.
Send orders to kitchen staff	VA	Necessary to initiate the preparation process.
Check orders on kitchen screens	VA	Helps kitchen staff track and prepare correct orders.
Prepare order	VA	Core task that fulfills the customer's request.
Tell manager that preparation is complete	BVA	Adds operational value but could be automated to reduce delays.
Send order to packager	VA	Ensures proper handover for packaging.
Package the order	VA	Guarantees items are securely packed and ready for delivery or pickup.
Check if the order is for takeaway/delivery	VA	Critical for routing the order to the correct workflow.
Define routes and couriers	NVA	A manual task that delays delivery and could be automated with routing software.
Provide TPA to courier	NVA	Administrative task that adds no direct value to the customer
Deliver the order	VA	Essential step that fulfills customer expectations.
Handle order issues or mistakes	NVA	A rework caused by errors earlier in the process, leading to unnecessary delays.
Offer discount or compensation for mistakes	NVA	A reactive measure caused by avoidable issues, adding no value to the process.
Collect payment	VA	Necessary to complete the transaction and provide closure for the customer.
Confirm VAT number	BVA	Regulatory compliance but not directly customer-focused.
Return incorrect orders to the store	NVA	Avoidable waste caused by insufficient checks earlier in the workflow.

Waste Analysis

1. Move

Transportation:

Lack of an automated routing system for delivery results in inefficient courier paths, increasing delivery times and operational costs.

Motion:

Manual handoffs between staff, such as transferring orders from the receptionist to the kitchen then between the managers/packager/courier, create unnecessary tasks.

2. Hold

Waiting:

High resource utilization in bottlenecked roles (e.g., kitchen staff and managers) creates delays for dependent tasks.

Customers experience waiting times during peak hours due to insufficient staffing and manual task processing.

3. Overdo

Defects:

Errors in order preparation and delivery lead to rework, including returning incorrect orders and compensating customers.

Overprocessing:

Double verification of VAT numbers and manual data entry for orders are redundant and add no direct value to the customer.

Issue Register

The Issue Register is a crucial tool for identifying and documenting the challenges and inefficiencies within the Order-to-Cash (O2C) process at Home Burger. By capturing these issues, we can address areas that hinder process flow, increase costs, or negatively affect customer satisfaction.

Each identified issue is categorized based on its impact, whether it relates to non-value-adding activities, bottlenecks, delays, or errors.

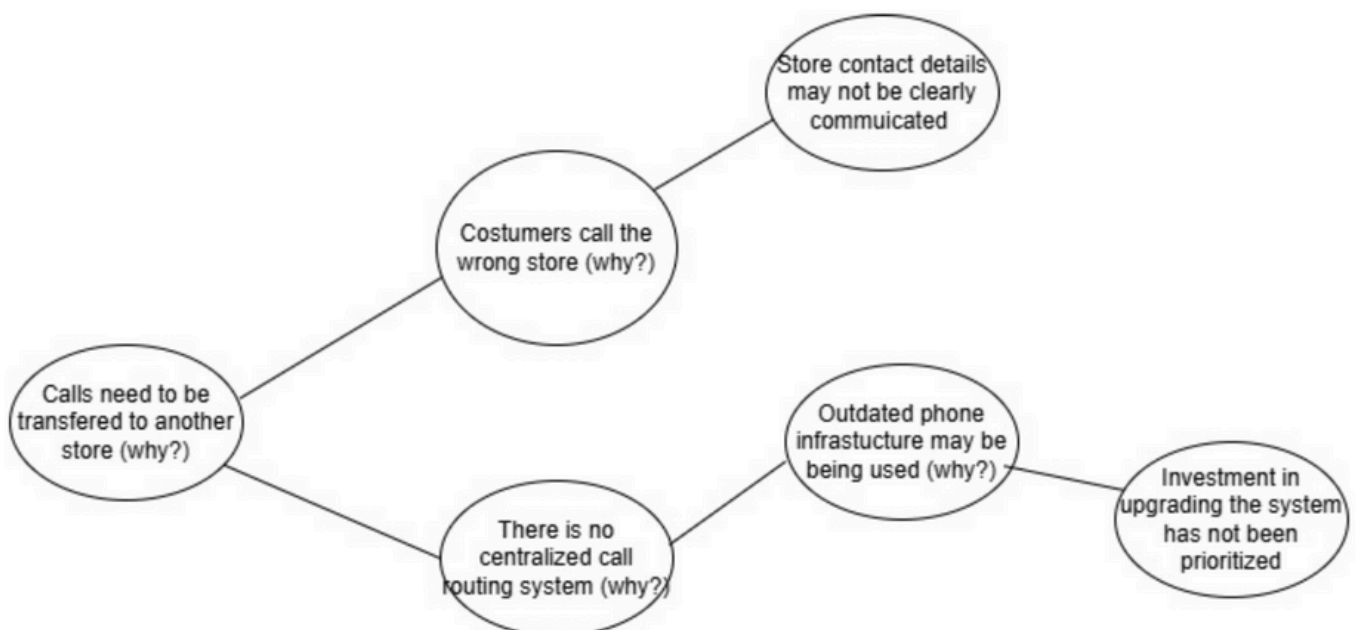
ISSUE	ROOT CAUSE	IMPACT	PRIORITY
Returning Incorrect Orders	Lack of verification	Customer dissatisfaction, delivery delays, increased costs	High
Manual Route Assignment	No route optimization system	Delivery inefficiencies, increased operational costs	Medium
Transferring calls to other stores	No centralize call-routing system	Wasted time, customer frustration, increased workload	Medium
Manual Checking of Website Orders	Reliance of manual workflows	Inefficiency, errors in order preparation	High
Calls not answered	Insufficient staffing or system overload	Missed orders, customer frustration, lost revenue	High
Downloading Excel Sheets	Data integration between systems eliminate the need for manual downloads is not implemented	Inefficiency, delays in order processing	High
Manual Messaging from Courier to Manager	No automated communication channel between couriers and managers	Delays in reporting delivery issues, inefficiency	Low

Why why analysis

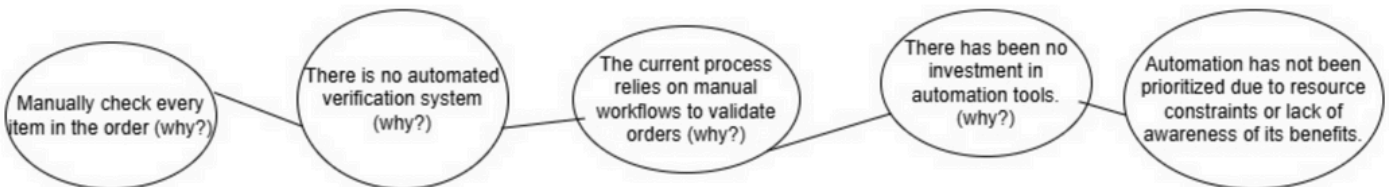
The Why-Why Analysis is a problem-solving tool used to identify the root causes of inefficiencies and issues within the Order-to-Cash (O2C) process at Home Burger. By repeatedly asking "Why?" for each identified problem, we can trace its origins and uncover deeper, systemic causes that might otherwise go unnoticed.

This method helps move beyond surface-level symptoms and provides actionable insights for addressing challenges such as delays, errors, and non-value-adding activities. Each diagram represents a structured breakdown of an issue, leading to the identification of root causes and opportunities for improvement. These insights will inform the redesign of the O2C process to improve efficiency, reduce costs, and enhance customer satisfaction.

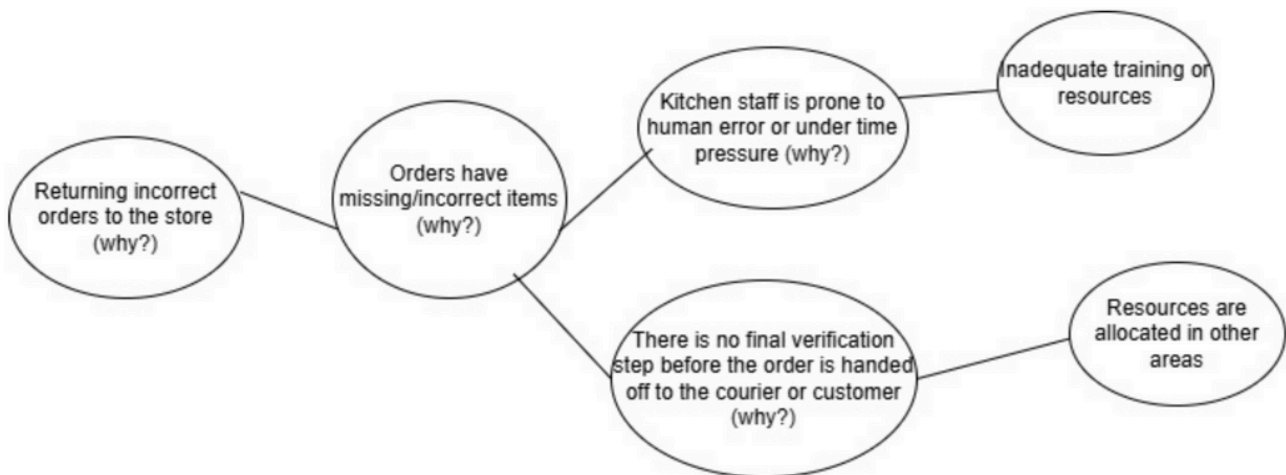
Orders registered by phone analysis



Orders registered by the website analysis



Orders processing analysis



Quantitative Analysis

The Quantitative Analysis evaluates the Order-to-Cash (O2C) process at Home Burger using measurable data to identify inefficiencies and assess performance. By analyzing metrics such as process times, waiting times, resource utilization, and costs, this section provides a detailed understanding of how effectively the current process operates.

The analysis includes key performance indicators such as average processing times, bottlenecks, and task completion rates. Simulation results are also incorporated to evaluate resource allocation, identify areas of overutilization or underutilization, and test potential improvements.

Orders registered by phone

The company registered a total of 5250 orders. Out of these, 52 percent were placed through the call center, amounting to 2730 orders. On average, this translates to 91 orders per day registered at the call center over a 30-day period.

However, only 70 percent of calls to the call center are answered. To calculate the total number of calls received, divide the average daily orders by the call answer rate:

$$\text{Total calls received per day} = 91 / 0.7 = 130$$

The start event of the call process follows a Poisson distribution. Given an 8-hour workday (equivalent to 28800 seconds), the mean arrival rate (λ) is calculated as:

$$\text{Mean arrival rate} = 130 / 28800 = 0.00451 \text{ calls per second}$$

The mean inter-arrival time, representing the average time between two consecutive calls, is the reciprocal of λ :

$$\text{Mean inter-arrival time} = 1 / 0.00451 = 221.73 \text{ seconds}$$

This means, on **average**, a call is received every **221.73 seconds**.

Cost

The monthly salary for a full-time position is 1425 euros. Assuming the employee works 8 hours per day, 7 days a week, over 4 weeks in a month, the total number of working hours is 224 (calculated as 8 hours per day multiplied by 7 days per week, then multiplied by 4 weeks).

Dividing the monthly salary by the total number of hours worked:

$$1425 \text{ euros} / 224 \text{ hours} = 6.36 \text{ euros per hour.}$$

Therefore, the **hourly rate** for a full-time employee working 7 days a week is approximately **6.36 euros**.

AS-IS simulation

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Phone operator	30.21 %	0	45.67	45.67

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting	Max. time waiting	Avg. time waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	442.64	198.85	25850.96			
Enter customers phone number into the order IS	Task	16.40	24.96	20.14	1792.33	0	0	0
Verify if customer made an order before	Task	25.02	34.12	29.62	2636.57	0	0	0
View customer data for delivery	Task	40.17	59	49.58	4412.8	0	0	0
Ask for customer information	Task	18.19	24.03	20.76	394.47	0	0	0
Register information in the database	Task	90.02	118.56	103.52	1966.89	0	0	0
Order fulfillment	Task	170.15	189.65	178.92	12703.16	0	0	0

Complete table: Annex 1.

In the AS-IS scenario for the "Order-to-Cash" process, the average process time was 198.85 seconds (approximately 3.31 minutes). This includes all tasks, from the customer placing the call to the completion of the order. The phone operator's utilization was 30.21%, indicating that they are handling the workload efficiently without being overburdened.

The most time-intensive tasks in the process were Order Fulfillment, taking an average of 178.92 seconds (around 2.98 minutes), and Registering Information in the Database, which averaged 103.52 seconds (approximately 1.72 minutes). These tasks represent significant portions of the overall process time.

Despite these time-intensive steps, the simulation shows minimal or no waiting times for tasks, suggesting that the current setup with one phone operator effectively meets the demand. However, opportunities exist to optimize tasks like database entry and fulfillment to further reduce the overall process time.

What if scenario 1

In this scenario we changed the number of website operators from 3 to 2.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Phone operator	45.31 %	0	45.67	45.67

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time for waiting	Max. time for waiting	Avg. time for waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	442.64	198.99	25868.25			
Enter customers phone number into the order IS	Task	16.40	34.59	20.33	1809.62	0	16.65	0.19
Verify if customer made an order before	Task	25.02	34.12	29.62	2636.57	0	0	0
View customer data for delivery	Task	40.17	59	49.58	4412.80	0	0	0
Ask for costumer information	Task	18.19	24.03	20.76	394.47	0	0	0
Register information in the database	Task	90.02	118.56	103.52	1966.89	0	0	0
Order fulfillment	Task	170.15	189.65	178.92	12703.16	0	0	0

Complete table: Annex 2.

In Scenario 1, with two employees handling phone calls instead of three, the average processing time slightly increased to 198.99 seconds (approximately 3.32 minutes) compared to 198.85 seconds (around 3.31 minutes) in the AS-IS scenario. This minor increase of 0.14 seconds per order is negligible and does not significantly impact overall efficiency.

The most time-consuming tasks remained unchanged:

- Registering Information in the Database: 103.52 seconds (approximately 1.73 minutes).
- Order Fulfillment: 178.92 seconds (around 2.98 minutes).

Despite the reduction in staff, the system maintained minimal waiting times, with no significant delays observed.

In terms of resource utilization, the phone operators' utilization increased from 30.21% in the AS-IS scenario to 45.31% in Scenario 1, reflecting a higher workload for the remaining employees. However, this increased workload did not compromise process efficiency, making this scenario a cost-effective solution with optimized resource use.

What if scenario 2

In this scenario we changed the number of website operators from 3 to 1. Lowering the number of employees to 2 would be worth it for the company, so we wanted to see if lowering it down to 1 would further benefit.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Phone operator	90.32 %	0	45.67	45.67

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time for waiting	Max. time for waiting	Avg. time for waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	1188.52	375.68	48838.35			
Enter customers phone number into the order IS	Task	18.22	391.55	100.13	8912.35	0	372.20	80.00
Verify if customer made an order before	Task	25.26	376.46	84.92	7557.91	0	349.01	55.30
View customer data for delivery	Task	40.17	265.91	92.02	8190.03	0	214.34	42.44
Verify if it is a customer with a different number	Task	20.75	378.52	67.61	2974.80	0	354.33	42.83
Transfer the call to the right store	Task	16.42	206.59	68.22	1228.04	0	187.78	48.12
Register information in the database	Task	90.02	171.52	124.71	2369.39	0	68.08	21.18
Order fulfillment	Task	170.15	396.57	219.20	15562.96	0	217.45	40.28

Complete table: Annex 3.

In Scenario 2, where the number of website operators was reduced to one, the process experienced significant delays and inefficiencies. The average processing time increased to 375.68 seconds (6 minutes and 16 seconds), significantly higher than in scenarios with more operators. Total waiting time also escalated to 22,987.39 seconds (6 hours, 23 minutes, and 7 seconds), indicating severe bottlenecks.

Key observations include:

- "Enter customer phone number into the orders IS": Average task time was 100.13 seconds (1 minute and 40 seconds), with a total waiting time of 7,120.01 seconds (1 hour, 58 minutes, and 40 seconds).
- "Verify if it is a customer with a different number": Average time of 67.60 seconds (1 minute and 7 seconds) and total waiting time of 2,974.80 seconds (49 minutes and 34 seconds).
- "Transfer the call to the right store": Average waiting time increased to 48.11 seconds (48 seconds), with a total waiting time of 866.08 seconds (14 minutes and 26 seconds).
- Order Fulfillment: Waiting time alone averaged 219.16 seconds (3 minutes and 39 seconds), indicating delays throughout the end-to-end process.

While reducing to one operator cuts costs, the sharp rise in waiting times and inefficiencies makes this scenario unsustainable. These delays would result in poor customer experiences, unfulfilled demand, and potential financial losses, outweighing the cost-saving benefits. Maintaining a minimum of two operators is crucial for balancing efficiency and cost-effectiveness.

What if scenario 3

Since lowering the number of employees to 2 was beneficial and to 1 was not good, we wanted to see if they could handle more calls, so in this scenario we used 2 employees and increased the chance of a call being picked up to 90%.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Phone operator	57.75 %	0	58.15	58.15

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time for waiting	Max. time for waiting	Avg. time for waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	442.63	253.48	32953.29			
Enter customers phone number into the order IS	Task	15.21	34.58	20.38	2303.68	0	16.64	0.23
Verify if customer made an order before	Task	25.02	34.56	29.81	3368.97	0	0	0
View customer data for delivery	Task	40.16	59.0	49.49	5592.68	0	0	0
Ask for costumer information	Task	18.19	24.02	20.84	416.83	0	0	0
Transfer the call to the right store	Task	16.42	30.30	20.49	40.9.80	0	11.12	0.55
Register information in the database	Task	90.02	118.55	103.56	2071.28	0	0	0
Order fulfillment	Task	170.15	189.64	179.55	16698.64	0	0	0

Complete table: Annex 4.

In Scenario 3, with two employees and a 90% call pickup rate, the average processing time for the Order-to-Cash (O2C) process increased to 253.48 seconds (4 minutes and 13 seconds), compared to 198.85 seconds (3 minutes and 19 seconds) in the AS-IS scenario. This increase of 54.63 seconds reflects the additional load of calls being handled due to the higher pickup rate.

Key observations include:

- **Waiting Time Introduced:** The system now experiences an average waiting time of 26.35 seconds, indicating that some calls are placed on hold before being attended to. For instance:

"Enter customer phone number into the orders IS" saw a waiting time of 16.64 seconds.

"Transfer the call to the right store" experienced a waiting time of 11.12 seconds.

- **Increased Utilization:** Employee utilization rose to 57.75%, which is a significant increase compared to the AS-IS scenario, but still manageable without overburdening the staff.

While this configuration reduces the chance of missed calls and increases the system's ability to handle more requests, it introduces delays in processing and waiting times, making it less efficient overall. The trade-off between a higher call pickup rate and increased processing time suggests that further optimization may be required. Balancing employee resources with call pickup rates is essential for achieving better efficiency and maintaining customer satisfaction.

Orders registered by the website

The company registered a total of 5250 orders. Out of these, 25.5 percent were placed through the website, amounting to 1340 orders. On average, this translates to 45 orders per day registered on the website over a 30-day period.

The website operates for 5 hours per day, which is equivalent to 18,000 seconds (5 hours x 60 minutes x 60 seconds). To calculate the mean arrival rate (λ), we divide the total daily orders by the total number of seconds:

Mean arrival rate (λ) = 45 orders / 18,000 s = 0.0025 orders per second

The mean inter-arrival time, representing the average time between two consecutive orders, is the reciprocal of λ :

Mean inter-arrival time = 1 / 0.0025 = 400 seconds

This means, on **average**, a website order is placed every **400 seconds** (approximately 6.67 minutes).

Cost

The monthly salary for the website employee is 1500 euros. The employee works 5 hours per day, 7 days a week, over 4 weeks in a month, the total number of working hours is 140 (calculated as 5 hours per day multiplied by 7 days per week, then multiplied by 4 weeks).

Dividing the monthly salary by the total number of hours worked:

1500 euros / 140 hours = 10.71 euros per hour.

Therefore, the **hourly rate** for the website employee working 7 days a week is approximately **10.71 euros per hour**.

AS-IS Scenario

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Website Employee	48.70 %	0	25.84	25.84

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	130.08	227.74	193.03	8686.56			
Download spreadsheet with orders' details	Task	25.56	34.55	30.59	1376.54	0	0	0
Import order information into system database	Task	33.41	69.50	47.85	2153.33	0	0	0
Check customer's address	Task	16.07	23.05	20.17	907.43	0	0	0
Process order	Task	25.12	34.20	29.60	1095.19	0	0	0
Check if the order can go to another store	Task	28.89	38.02	33.78	270.26	0	0	0
Cancel order and invite to visit other store	Task	19.27	21.77	20.97	125.81	0	0	0
Manually check every item of the order	Task	54.74	89.23	73.95	2736.06	0	0	0

Complete table: Annex 5.

In the AS-IS scenario for the website, a single employee manages the entire process, resulting in an average processing time of 193.03 seconds (approximately 3.22 minutes). This duration reflects the time required to complete an order, starting from accessing the backend system to finalizing the order.

The most time-consuming tasks include:

- Manually Checking Orders: Averaging 73.95 seconds (approximately 1.23 minutes).
- Importing Order Information into the Database: Averaging 47.85 seconds (approximately 0.80 minutes).

The employee utilization rate was 48.7%, suggesting sufficient capacity to handle additional workload without introducing delays. Additionally, no waiting times were recorded for this process, ensuring smooth and uninterrupted task execution. This indicates that the current setup is efficient and capable of meeting demand under the AS-IS configuration.

What if scenario 1

In this scenario we wanted to see if the one employee could handle if Home Burguer started having more costumers through the website, so we increased the maximum arrival count by 50%, from 45 to 68.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Website Employee	77.26 %	0	39.14	39.14

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	130.08	237.47	193.46	13155.54			
Download spreadsheet with orders' details	Task	25.32	34.78	30.57	2078.91	0	0	0
Import order information into system database	Task	33.14	69.50	49.48	3364.62	0	0	0
Check costumers' address	Task	16.07	24.17	19.93	1355.20	0	0	0
Process order	Task	25.12	34.20	29.80	1668.86	0	0	0
Check if the order can go to another store	Task	28.89	39.61	34.47	413.60	0	0	0
Cancel order and invite to visit other store	Task	19.27	21.77	20.70	186.32	0	0	0
Manually check every item of the order	Task	53.23	89.23	72.45	4057.17	0	0	0

Complete table: Annex 6.

In this "What-If" scenario, the maximum customer arrivals through the website increased by 50%, from 45 to 68, to test whether a single employee could handle the additional workload. The results showed:

- The average process time remained consistent at 193.46 seconds (approximately 3.22 minutes), similar to the AS-IS scenario's 193.03 seconds, indicating no significant impact on the time taken to process individual orders.
- The total process time, however, rose significantly from 8,686.56 seconds (144.78 minutes) in the AS-IS scenario to 13,155.54 seconds (219.26 minutes), reflecting the increased workload.

The employee's utilization rate also increased from 48.7% in the AS-IS scenario to 77.26%, highlighting a much heavier workload. While the system still operates without delays or waiting times, this high utilization indicates the employee is nearing capacity.

If demand continues to rise or unexpected issues occur, this setup may lead to delays or inefficiencies. Further adjustments, such as adding more staff or streamlining tasks, may be needed to sustain higher workloads effectively.

What if scenario 2

In this scenario we kept the maximum arrival count of 68 (same as on the what if scenario 1) but we increased the number of website employees to 2.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Website Employee	38.63 %	0	39.14	39.14

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	130.08	237.47	193.46	13155.54			
Download spreadsheet with orders' details	Task	25.32	34.78	30.57	2078.91	0	0	0
Import order information into system database	Task	33.14	69.50	49.48	3364.62	0	0	0
Check customer's address	Task	16.07	24.17	19.93	1355.20	0	0	0
Process order	Task	25.12	34.20	29.80	1668.86	0	0	0
Check if the order can go to another store	Task	28.89	39.61	34.47	413.60	0	0	0
Cancel order and invite to visit other store	Task	19.27	21.77	20.70	186.32	0	0	0
Manually check every item of the order	Task	53.23	89.23	72.45	4057.17	0	0	0

Complete table: Annex 7.

In this scenario, the maximum customer arrival count through the website remained at 68, similar to Scenario 1, but the number of website employees was increased to 2. The results showed:

- The average process time stayed consistent at 193.46 seconds (approximately 3.22 minutes), the same as in Scenario 1. This indicates that increasing the number of employees did not impact the time taken to process individual orders.
- The total process time also remained at 13,155.54 seconds (approximately 219.26 minutes), as the workload distribution across two employees was smooth and efficient.

The utilization rate for each employee decreased significantly to 38.63%, compared to 77.26% in Scenario 1. This reduction in utilization highlights the improved workload distribution and reduced pressure on individual employees.

With no waiting times recorded, this setup ensures uninterrupted task execution, effectively handling the increased order volume. This configuration demonstrates an efficient balance between workload and employee capacity, providing flexibility to manage further increases in demand or handle unexpected delays without compromising performance.

Orders registered at the store

The company registered a total of 5250 orders, of which 22.5% were placed in person at the store, accounting for 1181 orders. On average, this translates to approximately 39 orders per day registered in the physical store over a 30-day period.

The store operates with one employee responsible for registering the in-person orders. Assuming the employee works for 8 hours per day, this is equivalent to 28,800 seconds (8 hours x 60 minutes x 60 seconds).

To calculate the mean arrival rate (λ), we divide the total daily orders by the total number of seconds the employee works per day:

Mean arrival rate (λ) = 39 orders / 28,800 s = 0.00135 orders per second.

The mean inter-arrival time, representing the average time between two consecutive orders, is the reciprocal of λ :

Mean inter-arrival time = 1 / 0.00135 = 740.74 seconds.

This means, on **average**, an in-person order is placed every **740.74 seconds** (approximately 12.68 minutes).

Cost

The monthly salary for a full-time position is 1500 euros. Assuming the employee works 8 hours per day, 7 days a week, over 4 weeks in a month, the total number of working hours is 224 (calculated as 8 hours per day multiplied by 7 days per week, then multiplied by 4 weeks).

Dividing the monthly salary by the total number of hours worked:

1500 euros / 224 hours = 6.70 euros per hour.

Therefore, the **hourly rate** for a full-time employee working 7 days a week is approximately **6.70 euros**.

AS-IS Scenario

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	35.79 %	0	18.81	18.81

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	209.17	377.21	259.21	10109.15			
Greet and ask if costumer is registered	Task	5.56	14.44	10.65	415.37	0	0	0
Ask if costumer wants to register	Task	5.65	14.94	9.79	195.78	0	0	0
Ask for information and register the costumer	Task	98.26	114.76	108.32	974.88	0	0	0
Ask for the order and answer any other questions	Task	160.19	199.38	178.50	6961.34	0	0	0
Ask for payment method	Task	25.25	34.20	29.91	1166.32	0	0	0
Confirm fiscal (VAT) number	Task	9.10	10.90	10.14	395.44	0	0	0

Complete table: Annex 8.

The process for handling in-person orders at the store was executed for a total of 39 instances. The average process time was 259.21 seconds (approximately 4.32 minutes), covering all tasks from greeting the customer to confirming the VAT number and finalizing the order.

Key observations:

- The most time-consuming task was "Ask for the order", averaging 178.50 seconds (approximately 2.98 minutes).
- Other tasks included "Confirm fiscal (VAT) number", with an average time of 10.14 seconds (0.17 minutes), and "Greet and ask if the customer is registered", averaging 10.65 seconds (0.18 minutes).

The total time spent fulfilling these 39 orders was 10,109.15 seconds (approximately 168.49 minutes). The in-store receptionist's utilization rate was 35.79%, indicating a moderate workload and sufficient capacity to handle additional tasks if required.

There were no recorded waiting times during this process, ensuring smooth execution and prompt customer service. The primary duration is attributed to collecting and confirming order details, which is critical for ensuring order accuracy and customer satisfaction.

What if scenario 1

Lowered the percentage of costumers that are registered from 45% to 20% and increased the percentage of costumers that want to register from 30% to 50%.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	39.29%	0	20.73	20.73

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	212.03	374.72	285.55	11136.48			
Greet and ask if costumer is registered	Task	5.56	14.44	10.65	415.37	0	0	0
Ask if costumer wants to register	Task	5.50	14.94	9.69	280.90	0	0	0
Ask for information and register the costumer	Task	98.26	114.76	106.51	1917.09	0	0	0
Ask for the order and answer any other questions	Task	160.19	199.38	178.50	6961.34	0	0	0
Ask for payment method	Task	25.25	34.20	29.91	1166.32	0	0	0
Confirm fiscal (VAT) number	Task	9.10	10.90	10.14	395.44	0	0	0

Complete table: Annex 9.

In this scenario, the percentage of customers already registered decreased to 20%, while the percentage of customers wanting to register increased to 50%. The process was executed for 39 instances, leading to an increase in the average process time to 285.55 seconds (approximately 4.76 minutes) compared to the AS-IS scenario.

Key observations with the most time-consuming tasks were:

- "Ask for information and register the customer", averaging 106.51 seconds (1.78 minutes).
- "Ask for the order and answer any other questions", averaging 178.50 seconds (2.98 minutes).

The total process time for fulfilling these 39 orders increased to 11,136.48 seconds (approximately 185.61 minutes). The in-store receptionist's utilization also rose to 39.29%, reflecting the increased workload.

Despite the longer process time, waiting times remained negligible, indicating that the system can handle the higher demand efficiently.

What if scenario 2

Kept the changes from scenario 1 and increased the max arrival count from 39 to 50.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	49.34 %	0	26.21	26.21

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	212.03	374.72	281.66	14082.91			
Greet and ask if costumer is registered	Task	5.56	14.66	10.73	536.41	0	0	0
Ask if costumer wants to register	Task	5.50	14.94	9.64	366.50	0	0	0
Ask for information and register the costumer	Task	98.26	114.76	105.92	2224.25	0	0	0
Ask for the order and answer any other questions	Task	160.19	199.38	179.35	8967.46	0	0	0
Ask for payment method	Task	25.25	34.75	29.69	1484.39	0	0	0
Confirm fiscal (VAT) number	Task	9.01	10.90	10.08	503.91	0	0	0

Complete table: Annex 10.

In this scenario, the maximum arrival count increased from 39 to 50, while retaining changes from Scenario 1. The average process time decreased slightly to 281.66 seconds (4.7 minutes) from 285.55 seconds (4.76 minutes) in Scenario 1, showing the system can handle more customers efficiently.

Key observations:

- "Ask for information and register the customer" averaged 105.92 seconds (1.76 minutes).
- "Ask for the order and answer any other questions" averaged 179.35 seconds (2.99 minutes).

The total process time increased to 14,082.91 seconds (234.71 minutes). The in-store receptionist's utilization rose to 49.34%, indicating a higher workload but still manageable.

This scenario demonstrates the system's scalability, maintaining smooth operations and service quality despite the increased customer volume.

Order Processing

The company registered a total of 5250 orders. Since all orders must go through the order processing system, this accounts for 100 percent of the total orders. On average, this translates to 175 orders per day processed by the system over a 30-day period.

The start event of the order processing workflow follows a Poisson distribution. Given an 8-hour workday (equivalent to 28800 seconds), the mean arrival rate (lambda) is calculated as:

Mean arrival rate = $175 / 28800 = 0.00608$ orders per second.

The mean inter-arrival time, representing the average time between two consecutive orders being processed, is the reciprocal of lambda:

Mean inter-arrival time = $1 / 0.00608 = 164.47$ seconds.

This means, on **average**, a new order enters the order processing system every **164.47 seconds**.

Cost

Assuming a full-time employee works 8 hours per day, 7 days a week, over 4 weeks (224 hours per month):

Reception employee:

Monthly Salary: 1500€

Hourly Rate: $1500\text{€} / 224 = 6.70\text{€}$ per hour

Kitchen Staff:

Monthly Salary: 1875€

Hourly Rate: $1875\text{€} / 224 = 8.37\text{€}$ per hour

Packagers:

Monthly Salary: 1500€

Hourly Rate: $1500\text{€} / 224 = 6.70\text{€}$ per hour

Couriers:

Monthly Salary: 1350€

Hourly Rate: $1350\text{€} / 224 = 6.03\text{€}$ per hour

As the guidelines mentioned no salary for the manager but attributed him/her tasks, we assumed a monthly salary of 2000€.

Manager

Monthly Salary: 2000€

Hourly Rate: $2000\text{€} / 224 = 8.93\text{€}$

AS-IS Scenario

Resource	Utilization	Total cost
In-store receptionist	65.52 %	2107.22
Kitchen staff	91.85 %	14760.86
Packager	83.88 %	2697.73
Courier	84.67 %	19606.50
Manager	92.10 %	3947.64

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order processing	Process	953.37	9315.74	2898.18	489017.98			
Check orders on kitchen screen	Task	27.12	486.71	89.61	18728.77	0.00	456.63	59.82
Send orders to kitchen staff	Task	27.01	118.46	38.62	6759.31	0.00	86.93	8.75
Order prepared	Task	390.58	852.02	490.02	99964.56	0.00	421.65	39.78
Tell manager that the order is complete and the order ID	Task	27.15	477.92	78.40	15916.19	0.00	446.49	48.38
Order sent to the packager	Task	27.29	323.41	143.86	29203.47	0.00	292.79	113.94
Order packaged	Task	108.29	443.78	186.13	37411.29	0.00	321.67	66.33
Order delivered to customer	Task	54.49	120.60	77.02	2156.66	0.00	61.72	17.79
Define routes and couriers	Task	80.21	318.25	181.10	30787.33	0.00	226.78	91.50
TPA provided to courier	Task	27.28	329.63	149.75	25307.22	0.00	296.88	119.74
Order delivered	Task	805.12	1131.69	921.00	150122.92	0.00	153.78	14.57
Returned to the store	Task	802.77	1228.63	924.15	26800.31	0.00	232.74	22.05

Complete table: Annex 11.

The average process time for the Order Processing workflow was 2898.18 seconds (48.3 minutes), with a maximum time of 9315.74 seconds (155.3 minutes).

Key observations:

- "Order Delivered" averaged 921.00 seconds (15.4 minutes), making it the most time-consuming task.
- "Order Prepared" averaged 490.02 seconds (8.2 minutes).
- "Order Packaged" was more efficient, averaging 186.13 seconds (3.1 minutes).

While the Delays were observed in:

- "Order Sent to the Packager", with an average waiting time of 113.94 seconds (1.9 minutes).
- "Define Routes and Couriers", with an average waiting time of 91.50 seconds (1.5 minutes).

Resource utilization highlighted potential bottlenecks, with kitchen staff operating at 91.85% and the manager at 92.10%, while couriers and the packager were utilized at 84.67% and 83.88% , respectively. The receptionist had a lower utilization rate of 65.52 %, indicating unused capacity.

This analysis highlights bottlenecks in kitchen preparation and delivery logistics, while the receptionist has unused capacity.

What-If Scenario 1

The kitchen staff had too high of utilization and could be the cause of a bottleneck so in this scenario we added 1 kitchen staff to make it a total of 5.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	65.92 %	0	2120.14	2120.14
Kitchen staff	74.64 %	0	14992.96	14992.96
Packager	85.57 %	0	2751.94	2751.94
Courier	85.75 %	0	19854.50	19854.50
Manager	93.55 %	0	4009.84	4009.84

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	953.37	9145.14	2863.68	482984.99			
Check orders on kitchen screen	Task	27.00	159.61	33.36	7072.47	0.00	131.78	3.59
Send orders to kitchen staff	Task	27.07	114.74	37.93	6637.56	0.00	86.93	8.05
Order prepared	Task	390.58	527.80	453.18	94261.83	0.00	38.40	2.41
Tell manager that the order is complete and the order ID	Task	27.01	188.97	32.96	6855.37	0.00	160.07	2.96
Order sent to the packager	Task	27.29	409.22	166.84	34536.63	0.00	376.90	136.93
Order packaged	Task	108.29	570.17	207.07	42449.81	0.00	453.13	87.00
Order delivered to customer	Task	54.49	118.39	74.07	2148.02	0.00	58.67	14.94
Define routes and couriers	Task	80.21	409.85	198.13	34275.99	0.00	322.12	109.33
TPA provided to courier	Task	27.28	414.17	171.56	29508.42	0.00	385.26	141.56
Order delivered	Task	805.12	1074.51	918.79	152519.73	0.00	96.59	12.57
Returned to the store	Task	802.77	1212.69	927.45	26896.12	0.00	302.48	25.14

Complete table: Annex 12.

In the As-Is scenario, the kitchen staff utilization was 91.85%, creating a bottleneck that impacted process efficiency. Adding one more kitchen staff member reduced utilization to 74.64%, alleviating pressure on this critical resource.

Key observations:

- Overall Process Time: Average time decreased slightly from 2898.18 seconds (48.3 minutes) in the As-Is scenario to 2863.68 seconds (47.7 minutes).
- Average time reduced from 490.02 seconds (8.2 minutes) to 453.18 seconds (7.6 minutes).
- Maximum waiting time dropped significantly from 421.65 seconds to 38.40 seconds, nearly eliminating resource allocation delays.

Improvements:

- Waiting times for kitchen-related tasks showed substantial reductions, streamlining operations and improving workflow fluidity.

Trade-offs:

- Average waiting time for the Order Sent to the Packager task increased from 113.94 seconds (1.9 minutes) to 136.93 seconds (2.3 minutes), highlighting a need for further optimization in task transitions.

The addition of one kitchen staff member significantly reduced task durations and minimized waiting times for critical tasks like Order Prepared. This change enhanced workflow efficiency, reduced potential bottlenecks, and provided enough capacity to handle peak demand effectively. While the improvement in overall process time was modest, the impact on specific tasks was substantial, improving overall performance.

What-If Scenario 2

Kept the changes from scenario 1. Furthermore, as certain manager tasks were causing the bottleneck now and the manager already had high utilization we made it so that the tasks on the manager lane could be done by the manager or the receptionist.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	92.33 %	0	2969.38	2969.38
Kitchen staff	74.64 %	0	14994.21	14994.21
Packager	86.35 %	0	2776.93	2776.93
Courier	86.94 %	0	20131.66	20131.66
Manager	69.07 %	0	2960.55	2960.55

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order processing	Process	966.81	10489.67	2545.85	434619.10			
Check orders on kitchen screen	Task	27.00	152.67	33.70	7145.08	0.00	124.11	3.93
Send orders to kitchen staff	Task	27.07	145.38	55.35	9687.07	0.00	117.86	25.48
Order prepared	Task	390.58	524.90	453.26	94731.90	0.00	56.76	2.54
Tell manager that the order is complete and the order ID	Task	27.01	159.94	33.01	6899.63	0.00	132.07	3.02
Order sent to the packager	Task	27.01	122.66	46.91	9803.92	0.00	93.73	16.99
Order packaged	Task	109.28	504.91	222.18	45990.42	0.00	385.32	101.78
Order delivered to customer	Task	54.72	155.69	95.90	2781.05	0.00	100.45	35.99
Define routes and couriers	Task	80.20	142.22	97.15	17196.26	0.00	53.63	7.17
TPA provided to courier	Task	27.13	123.94	36.95	6540.29	0.00	93.45	6.97
Order delivered	Task	810.64	1054.62	919.27	153517.88	0.00	149.97	17.39
Returned to the store	Task	802.77	1023.42	917.77	26615.40	0.00	144.58	22.69

Complete table: Annex 13.

The overall process time in Scenario 2 averaged 2545.85 seconds (42.4 minutes), a reduction from 2863.68 seconds (47.7 minutes) in Scenario 1 and 2898.18 seconds (48.3 minutes) in the As-Is scenario.

Key observations:

- Manager utilization decreased significantly from 93.55% in Scenario 1 to 69.07%, reducing strain on this role.
- Receptionist utilization increased from 65.92% to 92.33%, effectively utilizing available capacity.
- Other resources, such as kitchen staff (74.64 percent), packager (86.35 percent), and courier (86.94 percent), remained stable, indicating balanced workloads.

Task waiting times improved:

- Order Sent to the Packager: Average waiting time reduced from 136.93 seconds (2.3 minutes) in Scenario 1 to 16.99 seconds (0.28 minutes).
- Order Packaged: Waiting time dropped from 101.78 seconds to 35.99 seconds.
- Define Routes and Couriers: Waiting time decreased from 109.33 seconds to 7.17 seconds.

This scenario demonstrates the effectiveness of redistributing tasks to balance workloads, particularly reducing the manager's strain and maximizing receptionist capacity. The reductions in waiting times and overall process time highlight improved task transitions and workflow efficiency.

What-If Scenario 3

In this scenario we added 1 courier and 1 packager, comparing to the what if scenario 2 as they had the highest utilization.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	93.14 %	0	2995.53	2995.53
Kitchen staff	74.88 %	0	15042.89	15042.89
Packager	43.41 %	0	2792.02	2792.02
Courier	78.56 %	0	20465.33	20465.33
Manager	68.60 %	0	2940.64	2940.64

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	953.37	7517.87	2435.73	409260.46			
Check orders on kitchen screen	Task	27.00	222.09	33.19	7069.97	0.00	193.53	3.41
Send orders to kitchen staff	Task	27.07	205.88	56.41	9872.11	0.00	177.06	26.53
Order prepared	Task	390.58	667.90	453.01	94679.94	0.00	160.08	2.28
Tell manager that the order is complete and the order ID	Task	27.01	90.70	32.38	6767.23	0.00	62.39	2.39
Order sent to the packager	Task	27.03	119.91	48.12	10056.28	0.00	90.47	18.20
Order packaged	Task	108.27	199.60	123.26	25638.20	0.00	82.99	3.23
Order delivered to customer	Task	55.95	139.97	95.02	2850.47	0.00	82.95	35.95
Define routes and couriers	Task	80.20	172.68	100.04	17707.26	0.00	74.03	10.12
TPA provided to courier	Task	27.11	129.00	48.88	8651.42	0.00	98.37	18.90
Order delivered	Task	805.12	1038.85	910.11	154718.43	0.00	148.14	4.87
Returned to the store	Task	802.77	995.90	902.31	27971.48	0.00	30.69	3.04

Complete table: Annex 14.

The overall average process time in Scenario 3 was 2435.73 seconds (40.6 minutes), reduced from 2545.85 seconds (42.4 minutes) in Scenario 2, highlighting the positive impact of adding a courier and a packager.

Key observations:

- Average time decreased from 222.18 seconds (3.7 minutes) in Scenario 2 to 123.26 seconds (2.1 minutes).
- Waiting times reduced from 101.78 seconds to 82.99 seconds.

Resource utilization was better balanced, with courier utilization decreasing from 86.94% to 78.56% and packager utilization dropping from 85.57% to 43.41%, ensuring flexibility.

While the changes led to reduced task times and improved workflow efficiency, the additional staff significantly increased costs. This scenario demonstrates improved performance, but the cost implications suggest exploring alternative strategies for achieving similar results with lower expenses. Further analysis is recommended to evaluate cost-effectiveness.

What-If Scenario 4

In this scenario we kept building on the what-if scenario 3. However, we eliminated the manager and made it so that the packagers or the receptionist could do his/her tasks.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	91.60 %	0	2945.83	2945.83
Kitchen staff	77.58 %	0	15584.78	15584.78
Packager	82.44 %	0	5302.29	5302.29
Courier	82.66 %	0	21531.95	21531.95

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	922.26	10326.00	2712.32	456456.98			
Check orders on kitchen screen	Task	27.00	228.05	39.21	8625.88	0.00	198.12	9.41
Send orders to kitchen staff	Task	27.18	179.81	67.04	11731.17	0.00	151.05	37.16
Order prepared	Task	390.58	571.48	456.19	98993.21	0.00	61.96	5.19
Tell manager that the order is complete and the order ID	Task	27.01	253.92	38.39	8329.57	0.00	222.68	8.38
Order sent to the packager	Task	27.01	145.51	51.24	11119.06	0.00	114.94	21.31
Order packaged	Task	108.29	281.24	146.45	31633.42	0.00	149.34	26.48
Order delivered to customer	Task	57.28	236.83	112.63	4054.58	0.00	182.30	53.14
Define routes and couriers	Task	80.92	211.50	108.35	19394.32	0.00	121.50	18.19
TPA provided to courier	Task	27.16	138.03	53.20	9523.51	0.00	107.41	23.22
Order delivered	Task	805.12	1109.30	920.58	161101.43	0.00	152.34	15.21
Returned to the store	Task	802.77	1188.26	921.71	35946.75	0.00	214.31	16.98

Complete table: Annex 15.

The overall average process time in Scenario 4 increased slightly to 2712.32 seconds (45.2 minutes) from 2435.73 seconds (40.6 minutes) in Scenario 3. This reflects the impact of reallocating managerial tasks to packagers and the receptionist.

Key observations:

- Average time increased from 123.26 seconds (2.1 minutes) in Scenario 3 to 146.45 seconds (2.4 minutes).
- Average waiting time rose from 82.99 seconds to 149.34 seconds, indicating minor inefficiencies.
- Waiting time for Items Received and Left for Delivery increased from 37.19 seconds in Scenario 3 to 53.45 seconds.

Resource utilization rates showed mixed results. The packager utilization increased to 82.44 percent, while the courier utilization remained stable at 82.66 percent, indicating balanced workloads. The receptionist utilization increased slightly to 91.60 percent, reflecting their additional responsibilities. Kitchen staff utilization rose marginally to 77.58 percent, maintaining a manageable workload.

This scenario effectively eliminates the manager role, reducing costs. However, the redistribution of tasks caused minor inefficiencies, particularly in packaging and delivery tasks. This highlights the trade-off between cost savings and maintaining process efficiency.

What-If Scenario 5

In this scenario we wanted to explore the possibility that the orders coming back with mistakes were causing the problems. Moreover we considered the as-is scenario and decreased the chance of the orders having a mistake to 1%.

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
In-store receptionist	64.35 %	0	2069.52	2069.52
Kitchen staff	77.74 %	0	12493.19	12493.19
Packager	72.45 %	0	2329.92	2329.92
Courier	65.57 %	0	15183.30	15183.30
Manager	80.21 %	0	3438.11	3438.11

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order processing	Process	932.07	5013.59	1921.71	324231.24			
Check orders on kitchen screen	Task	27.02	101.18	30.89	5467.14	0.00	71.30	1.15
Send orders to kitchen staff	Task	27.01	117.13	36.70	6422.38	0.00	86.58	6.82
Order prepared	Task	390.58	532.53	451.10	78492.16	0.00	59.89	1.25
Tell manager that the order is complete and the order ID	Task	27.01	112.79	31.67	5511.30	0.00	83.97	1.66
Order sent to the packager	Task	27.01	205.01	71.18	12385.20	0.00	173.09	41.21
Order packaged	Task	108.29	229.49	133.18	23039.65	0.00	99.62	12.64
Order delivered to customer	Task	54.53	128.06	78.28	1800.50	0.00	69.00	18.08
Define routes and couriers	Task	80.88	227.46	113.29	16879.80	0.00	133.17	23.25
TPA provided to courier	Task	27.28	211.48	66.33	9816.95	0.00	182.27	36.25
Order delivered	Task	805.12	997.30	904.76	130284.84	0.00	2.07	0.01
Returned to the store	Task	877.71	903.20	890.45	1780.90	0.00	0.00	0.00

The overall average process time in Scenario 5 was 1921.71 seconds (32 minutes), a significant reduction from 2898.18 seconds (48.3 minutes) in the As-Is scenario. This improvement highlights the impact of reducing order errors to 1% on overall process performance.

Key observations:

- Average time reduced from 186.13 seconds (3.1 minutes) in the As-Is scenario to 133.18 seconds (2.2 minutes).

The reduction in errors also alleviated the workload for resources. The Packager utilization decreased from 83.88% in the As-Is scenario to 72.45%, and the Courier utilization dropped from 84.67% to 65.57%, providing additional flexibility during peak periods. The Manager utilization remained stable at 80.21%, reflecting consistent performance.

Tasks related to correcting errors, such as “Returned to the store,” were significantly minimized, improving order flow and reducing time spent on rework.

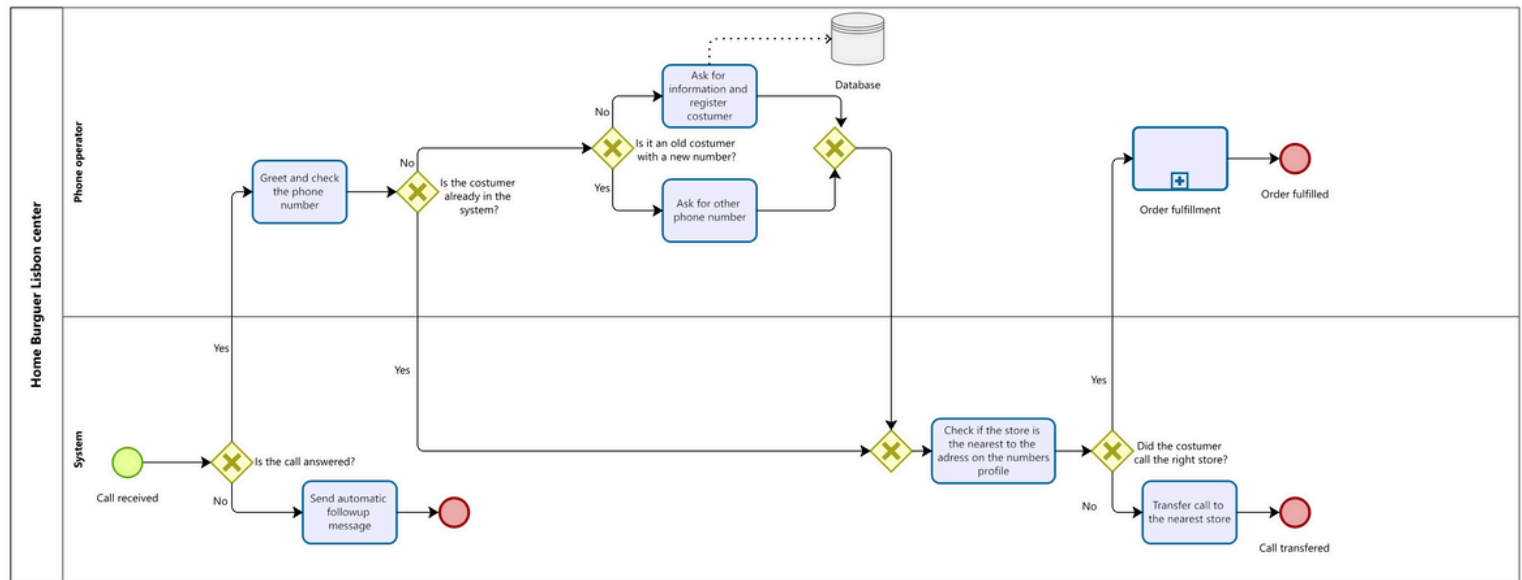
This scenario demonstrates the importance of quality control. Reducing the error rate led to shorter average process times, better resource balance, and fewer disruptions, resulting in a more efficient overall process.

To-Be Scenario

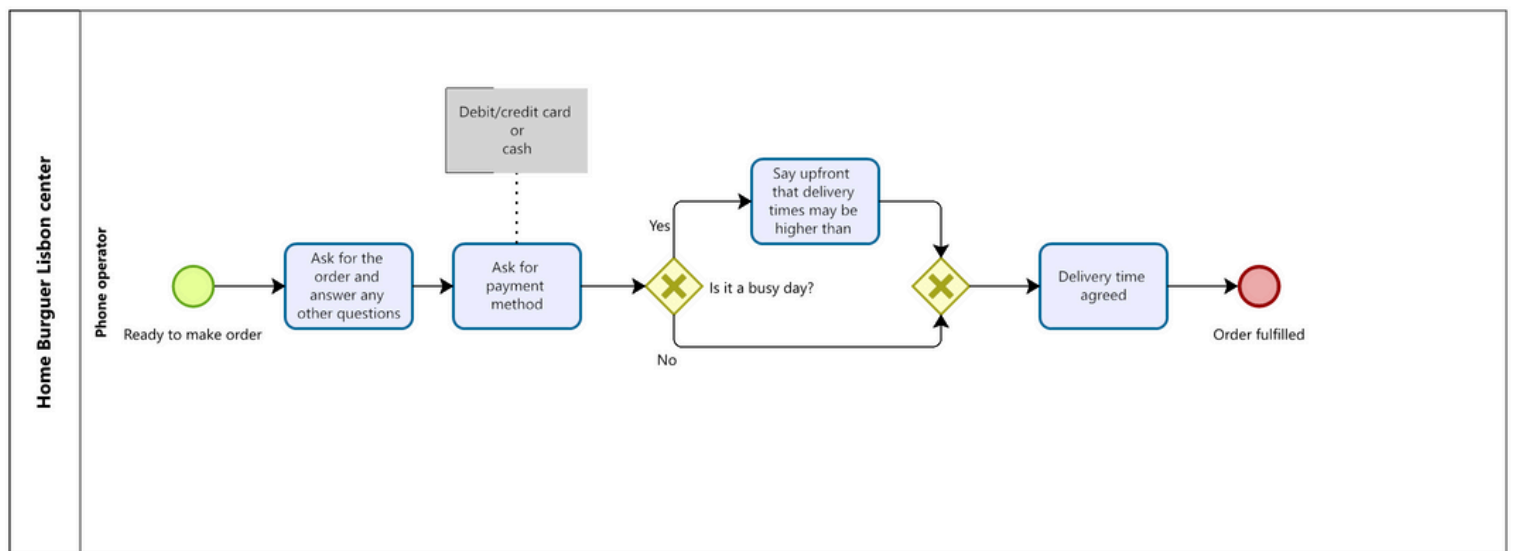
The proposed To-Be model for the Order-to-Cash process represents a significant step to improvement by addressing the inefficiencies and bottlenecks identified in the As-Is scenario. This model leverages modern technology and process redesign to streamline workflows, reduce manual effort, and enhance both customer satisfaction and resource utilization. By integrating automation, real-time data management, and improved communication systems, the To-Be model aims to optimize processes across all stages from order registration to delivery fulfillment. The goal is to create a more efficient, scalable, and cost-effective system that supports the business's growth and enhances its competitive edge.

To achieve this, the Toast POS system will be integrated as the backbone of the operations.

Orders registered by phone



Order fulfillment sub-process



In the To-Be process for phone orders, the integration of the Toast POS system revolutionizes how customer interactions and order management are conducted. When a call is received, the Toast system identifies whether the phone number is already registered in its database. If the caller is recognized, their profile and order history are automatically retrieved, enabling operators to handle the call with minimal input from the customer. This significantly reduces the time required for customer verification and enhances the overall experience.

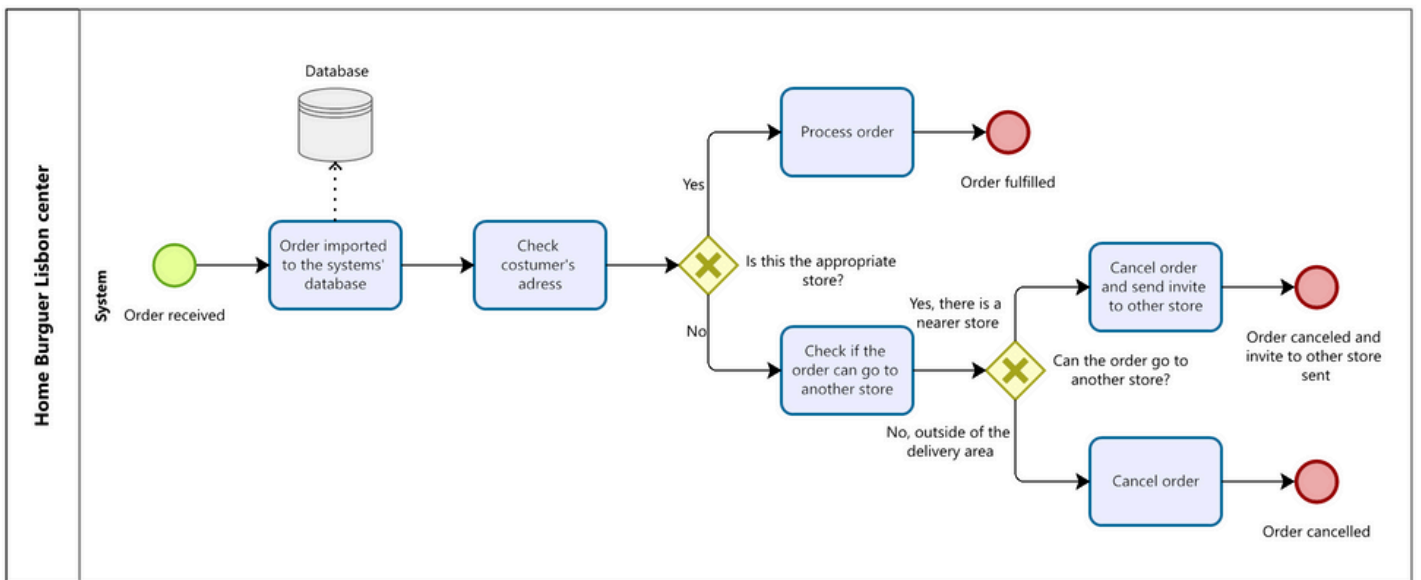
For new callers, the system prompts operators to collect and register essential information, including the customer's name, address, and contact details, ensuring a smooth onboarding process. For returning customers using a new phone number, Toast allows operators to merge the new details with an existing profile, maintaining a consistent and accurate database. This seamless integration eliminates data duplication and reduces errors in customer records.

A further enhancement is the direct connection of Toast with store-specific data, ensuring that customers are routed to the correct location. If a customer contacts a store that is not the closest to their address, the system automatically transfers the order or directs the call to the appropriate store. Additionally, for unanswered calls, Toast can be configured to trigger automated follow-up messages or emails, ensuring no customer inquiry goes unaddressed.

By adopting Toast POS, the organization implements a comprehensive and scalable solution that integrates phone orders, in-store operations, and delivery management into a single platform. The system eliminates manual inefficiencies, improves accuracy, and enhances customer satisfaction through its automation capabilities. The Toast Essentials Plan costs \$165/month for three phone operators and includes tools for caller identification, customer data retrieval, and efficient order processing.

The To-Be process also incorporates several key redesign heuristics to optimize operations. Elimination of manual data searches and repetitive tasks is achieved through Toast's automation features. Resource Optimization automates routine tasks, allowing operators to focus on value-added interactions with customers. Communication Optimization is achieved by enabling automated follow-ups for missed calls and seamless call routing to the correct store. These improvements collectively enhance operational efficiency and create a more consistent and satisfying customer experience.

Orders registered by website

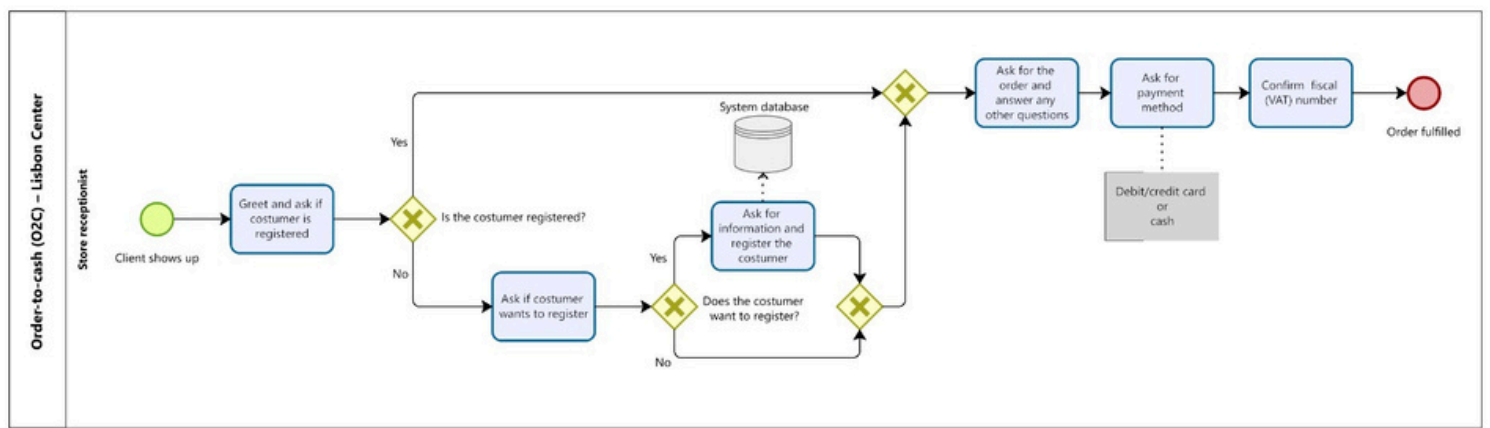


In the To-Be process for website orders, the integration of Toast POS enables a fully automated workflow, eliminating the need for any manual intervention. When an order is placed online, it is automatically imported into the system's centralized database. The system verifies the customer's address to ensure the order is assigned to the correct store. If the address aligns with the appropriate store, the order proceeds directly to fulfillment. If not, the system either redirects the order to the correct store or cancels it, notifying the customer in real-time to maintain transparency.

Automation within Toast POS ensures that all steps, from order placement to fulfillment, occur seamlessly. The system handles address verification, store assignment, and customer notifications without the need for manual input, significantly reducing processing times and errors. Customers receive real-time updates about their order status, improving satisfaction and trust in the process.

This fully automated approach eliminates the need for dedicated website employees and optimizes the order flow. By leveraging Toast POS's capabilities, the process applies key heuristics such as Automation for handling repetitive tasks, Communication Optimization through automated updates, and Resource Optimization by removing unnecessary human intervention. These enhancements ensure a streamlined, efficient, and customer-focused website order process.

Orders registered at the store

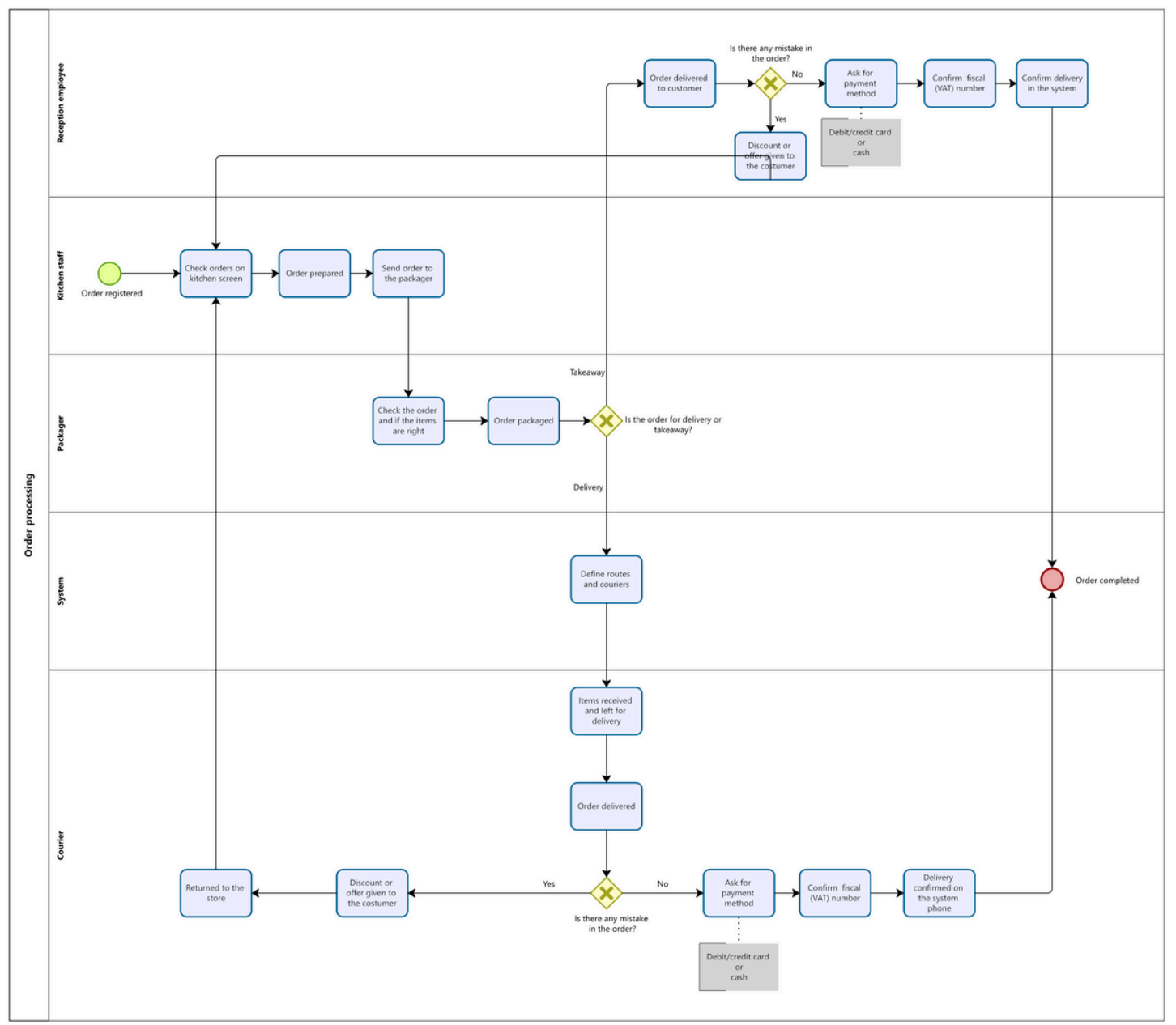


Orders Registered at the Store have been consistently efficient based on prior simulations and analysis. Hence we concluded that the Order Registered at the Store process does not require any changes in the To-Be model. After testing various scenarios in previous simulations, as detailed in the report, we observed that this process consistently performed efficiently across all situations. The simulations demonstrated that the process is already optimized in terms of task execution, resource utilization, and customer satisfaction.

In every scenario tested, the process showed no significant bottlenecks, delays, or errors that would necessitate further improvements. Additionally, the resource allocation was balanced, and the workflow structure supported seamless and accurate order registration. Based on these findings, we determined that the current state of the Order Registered at the Store process is well-suited to meet operational needs and does not require further adjustments in the To-Be model.

This allows us to focus on improving other areas of the order process, such as phone or online order registration, where optimization opportunities are more apparent.

Order Processing



In the To-Be process for order processing, Toast POS is integrated to streamline workflows, automate tasks, and enhance operational efficiency. Orders are automatically directed to the Kitchen Display System, eliminating the need for manual communication and ensuring that kitchen staff receives real-time updates. Delivery routing is also automated, leveraging tools like Google Maps to assign optimal routes for couriers based on their location and the delivery destination. This reduces delays and improves overall delivery accuracy.

Additionally, the manager role has been eliminated, with responsibilities redistributed to other staff and automated systems, ensuring a leaner and more efficient process. Dedicated smartphones for couriers, connected to the Toast POS system, provide real-time order updates, delivery instructions, and communication channels, ensuring seamless coordination between couriers and the store. Automating these tasks reduces errors, enhances delivery efficiency, and improves customer satisfaction.

This automated process eliminates the need for manual intervention in order management, routing, and communication, significantly reducing errors and processing times. Key heuristics such as Automation, Resource Optimization, and Communication Optimization are applied to streamline tasks, enhance collaboration between departments, and improve the customer experience. While the initial investment includes hardware and setup costs, the operational efficiencies and scalability provided by this system ensure long-term value for the business.

To-Be Quantitative Analysis

The To-Be process models represent a streamlined and automated approach to the Home Burger Lisbon Center's operations, addressing inefficiencies and resource bottlenecks identified in the As-Is analysis. By leveraging advanced systems such as Toast POS and automation through API integrations, the To-Be processes aim to enhance customer satisfaction, reduce processing times, and optimize resource utilization. This section provides a qualitative analysis of the proposed processes, highlighting improvements in workflow efficiency, task allocation, and overall operational effectiveness compared to the As-Is models.

Orders registered by phone

In the proposed scenario there is only one operator

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Phone Operator	74.19 %	0	37.75	37.75

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Home Burguer Lisbon center	Process	5.00	520.60	194.76	25152.57			
Greet and check the phone number	Task	15.32	246.56	34.68	4438.85	0.00	221.77	14.41
Check if the store is the nearest to the adress on the numbers profile	Task	10.00	10.00	10.00	1280.00	0.00	0.00	0.00
Ask for information and register costumers	Task	127.40	282.46	157.97	1895.66	0.00	146.29	24.09
Ask for other phone number	Task	8.05	11.91	9.79	450.18	0.00	0.00	0.00
Transfer call to the nearest store	Task	5.00	5.00	5.00	80.00	0.00	0.00	0.00
Send automatic followup message	Task	5.00	5.00	5.00	10.00	0.00	0.00	0.00
Order fulfillment	Task	140.90	286.88	153.13	16997.88	0.00	138.23	2.87

Complete table: Annex 17.

The To-Be phone order process achieved an average process time of 194.76 seconds (3.2 minutes), a slight improvement from 198 seconds in the previous To-Be scenario. Although the decrease is minimal, the process is now fully automated, significantly reducing manual intervention.

Key observations:

- Resource utilization for the single operator was 74.19%, indicating an effective workload distribution without overburdening staff.
- Key tasks:
 - "Greet and check the phone number" averaged 34.68 seconds.
 - "Order fulfillment" averaged 153.13 seconds, aided by automation.

Compared to the As-Is process, which relied heavily on manual intervention, the To-Be model eliminates inefficiencies, improves accuracy, and reduces errors. This automation ensures faster customer interactions, enhances performance, and lowers staffing costs, making it a cost-effective and efficient solution.

Orders registered by website

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Home Burguer Lisbon center	Process	48.58	62.03	55.71	2506.92			
Order imported to the systems' database	Task	18.45	21.85	20.00	900.18	0	0	0
Check costumer's address	Task	8.20	11.98	9.93	447.03	0	0	0
Process order	Task	20.76	29.88	25.13	929.89	0	0	0
Check if the order can go to another store	Task	18.30	20.92	19.76	158.08	0	0	0
Cancel order and send invite to other store	Task	8.29	11.85	10.41	62.47	0	0	0
Cancel order	Task	4.48	4.79	4.63	9.26	0	0	0

Complete table: Annex 18.

The To-Be website order process represents a significant improvement over the As-Is model, with the average process time reduced to 55.71 seconds (0.93 minutes) from 193.03 seconds (3.2 minutes) in the As-Is process, demonstrating a streamlined and automated workflow.

Key observations:

- Key tasks eliminated:
 "Manually check every item of the order".
 "Download spreadsheet with orders' details".

The removal of manual interventions significantly enhances accuracy and reduces errors.

Improvements:

- Automation: Shortens processing times and eliminates the need for a dedicated website employee, reducing costs while improving operational efficiency.

By automating these processes, the To-Be model not only shortens processing times but also eliminates the need for a dedicated website employee, cutting costs while improving operational efficiency. This approach ensures faster order fulfillment, enhanced customer satisfaction, and a scalable solution for future demands, making the To-Be process a clear advancement over the As-Is model.

Order Processing

Resource	Utilization	Total fixed cost	Total unit cost	Total cost
Reception employee	40.91 %	364	21.93	385.93
Kitchen staff	61.92 %	2112	207.30	2319.30
Packager	45.27 %	348	48.53	396.53
Courier	40.12 %	3144	135.47	3279.47

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	724.85	4347.57	1556.44	262556.82			
Send order to the packager	Task	27.04	32.93	30.20	5255.46	0.00	0.00	0.00
Check orders on kitchen screen	Task	27.13	32.99	29.95	5301.89	0.00	0.00	0.00
Check the order and if the items are right	Task	27.00	55.40	30.14	5244.19	0.00	23.15	0.18
Items received and left for delivery	Task	27.03	64.26	30.37	4372.56	0.00	33.52	0.31
Discount or offer given to the costumer	Task	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Confirm fiscal (VAT) number	Task	27.13	60.50	30.50	4208.46	0.00	30.74	0.43
Ask for payment method	Task	27.02	40.23	30.08	4151.10	0.00	10.35	0.15
Confirm fiscal (VAT) number	Task	27.10	32.91	30.12	843.39	0.00	0.00	0.00
Ask for payment method	Task	27.30	37.63	30.65	858.28	0.00	10.21	0.36
Discount or offer given to the costumer	Task	27.52	30.04	28.78	57.56	0.00	0.00	0.00
Returned to the store	Task	807.09	986.10	896.59	1793.19	0.00	0.00	0.00
Order delivered	Task	801.72	997.92	899.72	125961.46	0.00	31.15	0.42
Define routes and couriers	Task	13.05	15.95	14.72	2134.99	0.00	0.00	0.00
Delivery confirmed on the system phone	Task	9.02	62.36	10.42	1427.31	0.00	52.01	0.40
Order delivered to customer	Task	54.21	72.02	60.71	1699.77	0.00	11.19	0.64
Order prepared	Task	391.51	509.52	448.74	78080.30	0.00	0.00	0.00
Order packaged	Task	108.11	132.76	120.37	20823.23	0.00	1.02	0.01
Confirm delivery in the system	Task	9.10	71.04	12.27	343.69	0.00	60.83	2.17

In the new To-Be order processing model, the manager role has been eliminated, and tasks have been redistributed to create a streamlined and efficient workflow. The team now consists of :

1 reception employee
5 kitchen staff
2 packager
7 couriers

Key observations:

- Packagers are now responsible for verifying the accuracy of orders before dispatch, ensuring minimal order returns due to errors and eliminating bottlenecks.
- Automation: Tasks such as order registration and communication with the kitchen staff via the Kitchen Display System streamline operations.

Performance improvements:

- Average process time reduced from 2898 seconds (48.3 minutes) in the As-Is model to 1556 seconds (25.9 minutes) in the To-Be model.
- Waiting times have been virtually eliminated due to improved quality control and task automation.

This redistribution of tasks, combined with enhanced automation and quality control measures, ensures a more balanced workload for all roles while preventing delays and improving the overall efficiency of the process. These changes enable faster, more reliable service, creating a seamless experience for both staff and customers.

Financial Analysis

Cost of implementation

One-Time Costs:

- **Kitchen Display Systems:** 1,000€ (two screens at 500€ each).
- **Smartphones for Couriers:** 1,750€ (seven devices at 250€ each).
- **Training:** 500€ (for ten employees at 50€ each).

Total cost= 1,000+1,750+500 = 3,250€

Recurring Annual Costs:

- **Toast POS Subscription:** 6,000€ (at 500€ per month).
- **Mobile Plans for Couriers:** 1,680€ (seven devices at 20€ per month).
- **Delivery Routing API:** 60€.

Total Recurring Costs: 7,740€ annually.

Workforce costs

Role	As-Is Workers	As-Is Workforce Costs (€)	To-Be Workers	To-Be Workforce Costs (€)	
Phone Operators	3	51,300	1	17,100	
Reception Employee	1	18,000	1	18,000	
Kitchen Staff	4	90,000	5	112,500	
Packager	1	18,000	2	36,000	
Couriers	8	129,600	7	113,400	
Manager	1	24,000	0	0	
Total	18	330,900	16	296,000	- 34,900€ annually

Return on Investment (ROI)

The ROI can now be calculated based on the workforce cost savings (€34,900 annually) compared to the implementation cost (€10,990).

$$\text{ROI} = \frac{\text{Annual Savings} - \text{Recurring Costs}}{\text{One-Time Costs} + \text{Recurring Costs}} \times 100$$

$$\text{ROI} = \frac{34,900 - 7,740}{10,990} \times 100$$

$$\text{ROI} = \frac{27,160}{10,990} \times 100 \approx 247.1\%$$

The cost of implementing the new system, including one-time expenses and the first-year recurring costs, amounts to €10,990. This investment is offset by significant workforce cost savings of €34,900 annually due to optimized staff allocation and task automation. After accounting for the recurring costs, the net annual savings are €27,160, resulting in an impressive ROI of approximately 247.1%. This demonstrates the financial viability of the proposed changes, ensuring that the investment pays off in under one year while continuing to provide cost savings in subsequent years.

Limitations

While the redesigned processes and analyses provide a clear pathway to improving efficiency and reducing costs, several limitations should be considered when interpreting the results and implementing the proposed changes.

1. Data Limitations

The analysis relies on simulated scenarios and assumptions based on average operational data, such as employee salaries, task durations, and customer arrival rates. Any inaccuracies or variations in these inputs could affect the outcomes of the analysis. Additionally, some data, such as real-world system integration costs or unanticipated workflow interruptions, were not fully available and required estimation.

2. Assumptions in Costing and Workforce

The cost estimates for one-time and recurring expenses, such as Toast POS, mobile plans, and training, were based on current market rates. Actual costs may vary depending on vendor negotiations, regional factors, or future price changes. Similarly, workforce changes, such as the removal of the manager role and reallocation of tasks, were based on theoretical models and may not fully capture the complexities of real-world organizational adjustments.

3. System and Technology Limitations

Implementing new systems like Toast POS and delivery routing APIs may present unforeseen technical challenges, including integration issues with existing infrastructure or downtime during deployment. Additionally, training employees to effectively use these tools may require more time and resources than initially anticipated.

4. External Market Factors

External influences, such as market competition, inflation, or technological advancements, could affect the long-term viability of the proposed solutions. Ongoing monitoring and adaptability will be necessary to address these factors.

By acknowledging these limitations, the report emphasizes the need for a flexible and iterative approach to process improvements, ensuring that the proposed models remain relevant and effective under evolving conditions.

Conclusion

To remain competitive in the rapidly evolving food delivery market, Home Burger must address inefficiencies within its Order-to-Cash (O2C) process. The analysis conducted in this study has identified critical areas for improvement, including resource imbalances, manual task dependencies, and process bottlenecks. These issues not only affect operational efficiency but also have a direct impact on customer satisfaction and cost management.

Key findings indicate that high resource utilization in certain roles, such as kitchen staff and managers, creates bottlenecks that delay order processing and delivery. Manual workflows, including repetitive data entry and order verification tasks, further worsen inefficiencies, leading to avoidable errors and increased processing times.

To address these challenges, the proposed recommendations include automating redundant tasks, balancing workloads through resource reallocation, and refining task responsibilities across roles. Simulations of proposed changes demonstrate significant reductions in process time, resource strain, and error rates. These adjustments enable smoother workflow transitions, improved scalability, and enhanced customer experiences. By implementing these changes, Home Burger will improve its operational performance, reduce costs, and increase its capacity to manage growing demand effectively. These improvements will strengthen its competitive position, ensure long-term sustainability, and enhance its reputation as a leader in the food delivery industry.

Annex

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting	Max. time waiting	Avg. time waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	442.64	198.85	25850.96			
Enter customers phone number into the order IS	Task	16.40	24.96	20.14	1792.33	0	0	0
Verify if customer made an order before	Task	25.02	34.12	29.62	2636.57	0	0	0
View customer data for delivery	Task	40.17	59	49.58	4412.8	0	0	0
Verify if it is a customer with a different number	Task	20.75	29.47	24.78	1090.2	0	0	0
Ask for other phone number	Task	16.94	22.31	19.70	429.59	0	0	0
Ask for costumer information	Task	18.19	24.03	20.76	394.47	0	0	0
Transfer the call to the right store	Task	16.42	24.32	20.11	361.95	0	0	0
Register information in the database	Task	90.02	118.56	103.52	1966.89	0	0	0
Order fulfillment	Task	170.15	189.65	178.92	12703.16	0	0	0

Table 1: Phone AS-IS Simulation. Page 21

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time for waiting	Max. time for waiting	Avg. time for waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	442.64	198.99	25868.25			
Enter customers phone number into the order IS	Task	16.40	34.59	20.33	1809.62	0	16.65	0.19
Verify if customer made an order before	Task	25.02	34.12	29.62	2636.57	0	0	0
View customer data for delivery	Task	40.17	59	49.58	4412.80	0	0	0
Verify if it is a customer with a different number	Task	20.75	29.47	24.78	1090.20	0	0	0
Ask for other phone number	Task	16.94	22.31	19.70	492.59	0	0	0
Ask for costumer information	Task	18.19	24.03	20.76	394.47	0	0	0
Transfer the call to the right store	Task	16.42	24.32	20.11	361.95	0	0	0
Register information in the database	Task	90.02	118.56	103.52	1966.89	0	0	0
Order fulfillment	Task	170.15	189.65	178.92	12703.16	0	0	0

Table 2: Phone What if scenario 1. Page 23

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time for waiting	Max. time for waiting	Avg. time for waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	1188.52	375.68	48838.35			
Enter customers phone number into the order IS	Task	18.22	391.55	100.14	8912.35	0	372.20	80.00
Verify if customer made an order before	Task	25.26	376.46	84.92	7557.91	0	349.01	55.30
View customer data for delivery	Task	40.17	265.91	92.02	8190.03	0	214.34	42.44
Verify if it is a customer with a different number	Task	20.75	378.52	67.61	2974.80	0	354.33	42.83
Ask for other phone number	Task	17.53	234.55	55.35	1383.71	0	213.32	35.65
Ask for costumer information	Task	18.19	83.57	34.69	659.16	0	60.72	13.93
Transfer the call to the right store	Task	16.42	206.59	68.22	1228.04	0	187.78	48.12
Register information in the database	Task	90.02	171.52	124.71	2369.39	0	68.08	21.18
Order fulfillment	Task	170.15	396.57	219.20	15562.96	0	217.45	40.28

Table 3: Phone What if scenario 2. Page 25

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time for waiting	Max. time for waiting	Avg. time for waiting
Order-to-Cash (O2C) - Lisbon Center	Process	0	442.63	253.48	32953.29			
Enter customers phone number into the order IS	Task	15.21	34.58	20.38	2303.68	0	16.64	0.23
Verify if customer made an order before	Task	25.02	34.56	29.81	3368.97	0	0	0
View customer data for delivery	Task	40.16	59.0	49.49	5592.68	0	0	0
Verify if it is a customer with a different number	Task	20.72	29.47	24.81	1389.73	0	0	0
Ask for other phone number	Task	15.53	22.30	19.49	701.64	0	0	0
Ask for costumer information	Task	18.19	24.02	20.84	416.83	0	0	0
Transfer the call to the right store	Task	16.42	30.30	20.49	40.9.80	0	11.12	0.55
Register information in the database	Task	90.02	118.55	103.56	2071.28	0	0	0
Order fulfillment	Task	170.15	189.64	179.55	16698.64	0	0	0

Table 4: Phone What if scenario 3. Page 27

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	130.08	227.74	193.03	8686.56			
Download spreadsheet with orders' details	Task	25.56	34.55	30.59	1376.54	0	0	0
Import order information into system database	Task	33.41	69.50	47.85	2153.33	0	0	0
Check costumer's adress	Task	16.07	23.05	20.17	907.43	0	0	0
Process order	Task	25.12	34.20	29.60	1095.19	0	0	0
Check if the order can go to another store	Task	28.89	38.02	33.78	270.26	0	0	0
Cancel order and invite to visit other store	Task	19.27	21.77	20.97	125.81	0	0	0
Cancel order	Task	10.11	11.84	10.97	21.95	0	0	0
Manually check every item of the order	Task	54.74	89.23	73.95	2736.06	0	0	0

Table 5: Website AS-IS Simulation. Page 30

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	130.08	237.47	193.46	13155.54			
Download spreadsheet with orders' details	Task	25.32	34.78	30.57	2078.91	0	0	0
Import order information into system database	Task	33.14	69.50	49.48	3364.62	0	0	0
Check costumer's address	Task	16.07	24.17	19.93	1355.20	0	0	0
Process order	Task	25.12	34.20	29.80	1668.86	0	0	0
Check if the order can go to another store	Task	28.89	39.61	34.47	413.60	0	0	0
Cancel order and invite to visit other store	Task	19.27	21.77	20.70	186.32	0	0	0
Cancel order	Task	8.92	11.84	10.29	30.86	0	0	0
Manually check every item of the order	Task	53.23	89.23	72.45	4057.17	0	0	0

Table 6: Website What if scenario 1. Page 32

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	130.08	237.47	193.46	13155.54			
Download spreadsheet with orders' details	Task	25.32	34.78	30.57	2078.91	0	0	0
Import order information into system database	Task	33.14	69.50	49.48	3364.62	0	0	0
Check costumer's adress	Task	16.07	24.17	19.93	1355.20	0	0	0
Process order	Task	25.12	34.20	29.80	1668.86	0	0	0
Check if the order can go to another store	Task	28.89	39.61	34.47	413.60	0	0	0
Cancel order and invite to visit other store	Task	19.27	21.77	20.70	186.32	0	0	0
Cancel order	Task	8.92	11.84	10.29	30.86	0	0	0
Manually check every item of the order	Task	53.23	89.23	72.45	4057.17	0	0	0

Table 7: Website What if scenario 2.. Page 34

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	209.17	377.21	259.21	10109.15			
Greet and ask if costumer is registered	Task	5.56	14.44	10.65	415.37	0	0	0
Ask if costumer wants to register	Task	5.65	14.94	9.79	195.78	0	0	0
Ask for information and register the costumer	Task	98.26	114.76	108.32	974.88	0	0	0
Ask for the order and answer any other questions	Task	160.19	199.38	178.50	6961.34	0	0	0
Ask for payment method	Task	25.25	34.20	29.91	1166.32	0	0	0
Confirm fiscal (VAT) number	Task	9.10	10.90	10.14	395.44	0	0	0

Table 8: Store AS-IS Scenario. Page 36

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	212.03	374.72	285.55	11136.48			
Greet and ask if costumer is registered	Task	5.56	14.44	10.65	415.37	0	0	0
Ask if costumer wants to register	Task	5.50	14.94	9.69	280.90	0	0	0
Ask for information and register the costumer	Task	98.26	114.76	106.51	1917.09	0	0	0
Ask for the order and answer any other questions	Task	160.19	199.38	178.50	6961.34	0	0	0
Ask for payment method	Task	25.25	34.20	29.91	1166.32	0	0	0
Confirm fiscal (VAT) number	Task	9.10	10.90	10.14	395.44	0	0	0

Table 9: Store What if scenario 1. Page 38

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order-to-cash (O2C) – Lisbon Center	Process	212.03	374.72	281.66	14082.91			
Greet and ask if costumer is registered	Task	5.56	14.66	10.73	536.41	0	0	0
Ask if costumer wants to register	Task	5.50	14.94	9.64	366.50	0	0	0
Ask for information and register the costumer	Task	98.26	114.76	105.92	2224.25	0	0	0
Ask for the order and answer any other questions	Task	160.19	199.38	179.35	8967.46	0	0	0
Ask for payment method	Task	25.25	34.75	29.69	1484.39	0	0	0
Confirm fiscal (VAT) number	Task	9.01	10.90	10.08	503.91	0	0	0

Table 10: Store What if scenario 2. Page 40

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	953.37	9315.74	2898.18	489017.98			
Check system grid for order	Task	54.24	129.11	65.50	11462.66	0.00	65.86	5.26
Check orders on kitchen screen	Task	27.12	486.71	89.61	18728.77	0.00	456.63	59.82
Send orders to kitchen staff	Task	27.01	118.46	38.62	6759.31	0.00	86.93	8.75
Order prepared	Task	390.58	852.02	490.02	99964.56	0.00	421.65	39.78
Tell manager that the order is complete and the order ID	Task	27.15	477.92	78.40	15916.19	0.00	446.49	48.38
Order sent to the packager	Task	27.29	323.41	143.86	29203.47	0.00	292.79	113.94
Order packaged	Task	108.29	443.78	186.13	37411.29	0.00	321.67	66.33
Order delivered to customer	Task	54.49	120.60	77.02	2156.66	0.00	61.72	17.79
Define routes and couriers	Task	80.21	318.25	181.10	30787.33	0.00	226.78	91.50
TPA provided to courier	Task	27.28	329.63	149.75	25307.22	0.00	296.88	119.74
Order delivered	Task	805.12	1131.69	921.00	150122.92	0.00	153.78	14.57
Returned to the store	Task	802.77	1228.63	924.15	26800.31	0.00	232.74	22.05
Discount or offer given to the costumer	Task	27.07	178.42	45.47	1409.44	0.00	148.69	15.43
Ask for payment method	Task	28.59	124.11	67.37	1347.42	0.00	94.17	37.62
Confirm fiscal (VAT) number	Task	55.24	151.06	93.54	1870.87	27.50	120.44	63.55
Ask for payment method	Task	27.08	477.15	50.52	6669.21	0.00	448.58	20.63
Confirm fiscal (VAT) number	Task	27.52	260.72	51.80	6837.56	0.00	231.13	21.69
Delivery confirmed to the manager	Task	27.03	263.06	47.34	6249.25	0.00	234.94	17.32
Discount or offer given to the costumer	Task	29.84	153.24	68.58	548.62	0.00	122.31	37.69
Items received and left for delivery	Task	27.04	454.63	56.01	9464.91	0.00	424.81	25.93

Table 11: Order Processing AS-IS Scenario. Page 44

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	953.37	9145.14	2863.68	482984.99			
Check system grid for order	Task	54.24	153.02	67.42	11799.15	0.00	97.31	7.19
Check orders on kitchen screen	Task	27.00	159.61	33.36	7072.47	0.00	131.78	3.59
Send orders to kitchen staff	Task	27.07	114.74	37.93	6637.56	0.00	86.93	8.05
Order prepared	Task	390.58	527.80	453.18	94261.83	0.00	38.40	2.41
Tell manager that the order is complete and the order ID	Task	27.01	188.97	32.96	6855.37	0.00	160.07	2.96
Order sent to the packager	Task	27.29	409.22	166.84	34536.63	0.00	376.90	136.93
Order packaged	Task	108.29	570.17	207.07	42449.81	0.00	453.13	87.00
Order delivered to customer	Task	54.49	118.39	74.07	2148.02	0.00	58.67	14.94
Define routes and couriers	Task	80.21	409.85	198.13	34275.99	0.00	322.12	109.33
TPA provided to courier	Task	27.28	414.17	171.56	29508.42	0.00	385.26	141.56
Order delivered	Task	805.12	1074.51	918.79	152519.73	0.00	96.59	12.57
Returned to the store	Task	802.77	1212.69	927.45	26896.12	0.00	302.48	25.14
Discount or offer given to the costumer	Task	27.07	162.73	46.89	1453.46	0.00	132.99	16.85
Ask for payment method	Task	29.73	124.11	71.96	1511.26	0.00	94.17	42.22
Confirm fiscal (VAT) number	Task	32.33	124.32	81.30	1707.20	0.00	95.73	51.31
Ask for payment method	Task	27.08	442.21	48.43	6489.23	0.00	413.64	18.62
Confirm fiscal (VAT) number	Task	27.52	268.46	50.29	6688.36	0.00	235.66	20.25
Delivery confirmed to the manager	Task	27.03	236.46	46.57	6194.36	0.00	208.34	16.57
Discount or offer given to the costumer	Task	32.31	141.97	82.24	657.91	0.00	111.04	51.35
Items received and left for delivery	Task	27.04	398.91	54.20	9322.12	0.00	369.09	24.11

Table 12: Order Processing What if scenario 1. Page 47

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order processing	Process	966.81	10489.67	2545.85	434619.10			
Check system grid for order	Task	54.49	174.01	90.12	15770.52	0.00	117.08	29.88
Check orders on kitchen screen	Task	27.00	152.67	33.70	7145.08	0.00	124.11	3.93
Send orders to kitchen staff	Task	27.07	145.38	55.35	9687.07	0.00	117.86	25.48
Order prepared	Task	390.58	524.90	453.26	94731.90	0.00	56.76	2.54
Tell manager that the order is complete and the order ID	Task	27.01	159.94	33.01	6899.63	0.00	132.07	3.02
Order sent to the packager	Task	27.01	122.66	46.91	9803.92	0.00	93.73	16.99
Order packaged	Task	109.28	504.91	222.18	45990.42	0.00	385.32	101.78
Order delivered to customer	Task	54.72	155.69	95.90	2781.05	0.00	100.45	35.99
Define routes and couriers	Task	80.20	142.22	97.15	17196.26	0.00	53.63	7.17
TPA provided to courier	Task	27.13	123.94	36.95	6540.29	0.00	93.45	6.97
Order delivered	Task	810.64	1054.62	919.27	153517.88	0.00	149.97	17.39
Returned to the store	Task	802.77	1023.42	917.77	26615.40	0.00	144.58	22.69
Discount or offer given to the costumer	Task	27.07	154.89	50.03	1551.00	0.00	123.72	19.99
Ask for payment method	Task	27.86	125.88	84.74	1779.59	0.00	98.05	55.00
Confirm fiscal (VAT) number	Task	58.48	124.90	91.62	1924.03	28.27	95.74	61.64
Ask for payment method	Task	27.08	204.85	54.95	7472.55	0.00	177.05	25.04
Confirm fiscal (VAT) number	Task	27.52	281.69	59.38	7897.67	0.00	251.19	29.26
Delivery confirmed to the manager	Task	27.05	237.15	50.66	6687.19	0.00	204.38	20.64
Discount or offer given to the costumer	Task	29.88	89.71	64.38	515.07	0.00	59.82	33.49
Items received and left for delivery	Task	27.04	220.30	58.12	10112.55	0.00	189.87	28.03

Table 13: Order Processing What if scenario 2. Page 50

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	953.37	7517.87	2435.73	409260.46			
Check system grid for order	Task	54.39	191.26	93.75	16405.99	0.00	126.37	33.51
Check orders on kitchen screen	Task	27.00	222.09	33.19	7069.97	0.00	193.53	3.41
Send orders to kitchen staff	Task	27.07	205.88	56.41	9872.11	0.00	177.06	26.53
Order prepared	Task	390.58	667.90	453.01	94679.94	0.00	160.08	2.28
Tell manager that the order is complete and the order ID	Task	27.01	90.70	32.38	6767.23	0.00	62.39	2.39
Order sent to the packager	Task	27.03	119.91	48.12	10056.28	0.00	90.47	18.20
Order packaged	Task	108.27	199.60	123.26	25638.20	0.00	82.99	3.23
Order delivered to customer	Task	55.95	139.97	95.02	2850.47	0.00	82.95	35.95
Define routes and couriers	Task	80.20	172.68	100.04	17707.26	0.00	74.03	10.12
TPA provided to courier	Task	27.11	129.00	48.88	8651.42	0.00	98.37	18.90
Order delivered	Task	805.12	1038.85	910.11	154718.43	0.00	148.14	4.87
Returned to the store	Task	802.77	995.90	902.31	27971.48	0.00	30.69	3.04
Discount or offer given to the costumer	Task	27.07	89.27	34.49	1069.22	0.00	59.19	4.45
Ask for payment method	Task	57.11	146.11	82.36	1729.50	27.82	118.24	52.61
Confirm fiscal (VAT) number	Task	57.67	147.26	87.45	1836.46	27.60	114.93	57.47
Ask for payment method	Task	27.08	127.62	34.78	4835.00	0.00	95.83	4.87
Confirm fiscal (VAT) number	Task	27.48	220.50	39.40	5437.23	0.00	191.11	9.20
Delivery confirmed to the manager	Task	27.01	103.79	35.02	4832.24	0.00	76.24	5.03
Discount or offer given to the costumer	Task	31.67	123.67	68.69	549.50	0.00	93.79	37.20
Items received and left for delivery	Task	27.04	236.56	37.19	6582.52	0.00	207.17	7.08

Table 14: Order Processing What if scenario 3. Page 53

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Order processing	Process	922.26	10326.00	2712.32	456456.98			
Order registered	Start event							
Check system grid for order	Task	54.67	219.60	99.14	17349.78	0.00	155.55	38.90
Check orders on kitchen screen	Task	27.00	228.05	39.21	8625.88	0.00	198.12	9.41
Send orders to kitchen staff	Task	27.18	179.81	67.04	11731.17	0.00	151.05	37.16
Order prepared	Task	390.58	571.48	456.19	98993.21	0.00	61.96	5.19
Tell manager that the order is complete and the order ID	Task	27.01	253.92	38.39	8329.57	0.00	222.68	8.38
Order sent to the packager	Task	27.01	145.51	51.24	11119.06	0.00	114.94	21.31
Order packaged	Task	108.29	281.24	146.45	31633.42	0.00	149.34	26.48
Is the order for delivery or takeaway?	Gateway							
Order delivered to customer	Task	57.28	236.83	112.63	4054.58	0.00	182.30	53.14
Define routes and couriers	Task	80.92	211.50	108.35	19394.32	0.00	121.50	18.19
TPA provided to courier	Task	27.16	138.03	53.20	9523.51	0.00	107.41	23.22
Order delivered	Task	805.12	1109.30	920.58	161101.43	0.00	152.34	15.21
Is there any mistake in the order?	Gateway							
Returned to the store	Task	802.77	1188.26	921.71	35946.75	0.00	214.31	16.98
Is there any mistake in the order?	Gateway							
Order fulfilled	End event							
Discount or offer given to the costumer	Task	27.07	121.38	41.50	1618.58	0.00	93.12	11.50
Ask for payment method	Task	29.35	186.80	100.48	3014.36	0.00	156.63	70.76
Confirm fiscal (VAT) number	Task	58.48	204.39	106.46	3193.82	29.02	173.50	76.41
Ask for payment method	Task	27.08	294.03	47.85	6507.79	0.00	263.89	17.95
Confirm fiscal (VAT) number	Task	27.48	261.46	49.80	6772.77	0.00	230.43	19.67
Delivery confirmed to the manager	Task	27.30	290.59	52.84	7186.68	0.00	260.22	22.83
Delivery completed	End event							
Discount or offer given to the costumer	Task	93.65	242.91	132.16	792.94	62.72	210.59	101.37
Items received and left for delivery	Task	27.08	317.30	53.45	9567.35	0.00	286.21	23.33

Table 15:Order Processing What if scenario 4. Page 56

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting for resource (s)	Max. time waiting for resource (s)	Avg. time waiting for resource (s)
Order processing	Process	932.07	5013.59	1921.71	324231.24			
Check system grid for order	Task	54.39	136.46	65.46	11454.98	0.00	79.15	5.22
Check orders on kitchen screen	Task	27.02	101.18	30.89	5467.14	0.00	71.30	1.15
Send orders to kitchen staff	Task	27.01	117.13	36.70	6422.38	0.00	86.58	6.82
Order prepared	Task	390.58	532.53	451.10	78492.16	0.00	59.89	1.25
Tell manager that the order is complete and the order ID	Task	27.01	112.79	31.67	5511.30	0.00	83.97	1.66
Order sent to the packager	Task	27.01	205.01	71.18	12385.20	0.00	173.09	41.21
Order packaged	Task	108.29	229.49	133.18	23039.65	0.00	99.62	12.64
Order delivered to customer	Task	54.53	128.06	78.28	1800.50	0.00	69.00	18.08
Define routes and couriers	Task	80.88	227.46	113.29	16879.80	0.00	133.17	23.25
TPA provided to courier	Task	27.28	211.48	66.33	9816.95	0.00	182.27	36.25
Order delivered	Task	805.12	997.30	904.76	130284.84	0.00	2.07	0.01
Returned to the store	Task	877.71	903.20	890.45	1780.90	0.00	0.00	0.00
Discount or offer given to the costumer	Task	31.22	32.26	31.74	63.48	0.00	0.00	0.00
Ask for payment method	Task	27.84	122.42	67.90	1561.81	0.00	92.96	38.09
Confirm fiscal (VAT) number	Task	29.64	149.85	84.87	1951.93	0.00	120.02	54.79
Ask for payment method	Task	27.08	33.65	29.93	4250.23	0.00	0.72	0.01
Confirm fiscal (VAT) number	Task	27.48	59.49	30.35	4309.91	0.00	30.61	0.22
Delivery confirmed to the manager	Task	27.01	63.55	30.18	4285.61	0.00	30.87	0.22
Discount or offer given to the costumer	Task	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Items received and left for delivery	Task	27.04	56.31	30.22	4472.47	0.00	24.73	0.17

Table 16: Order Processing What if scenario 5. Page 59

Name	Type	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Home Burguer Lisbon center	Process	5.00	520.60	194.76	25152.57			
Greet and check the phone number	Task	15.32	246.56	34.68	4438.85	0.00	221.77	14.41
Check if the store is the nearest to the adress on the numbers profile	Task	10.00	10.00	10.00	1280.00	0.00	0.00	0.00
Ask for information and register costumer	Task	127.40	282.46	157.97	1895.66	0.00	146.29	24.09
Ask for other phone number	Task	8.05	11.91	9.79	450.18	0.00	0.00	0.00
Transfer call to the nearest store	Task	5.00	5.00	5.00	80.00	0.00	0.00	0.00
Send automatic followup message	Task	5.00	5.00	5.00	10.00	0.00	0.00	0.00
Order fulfillment	Task	140.90	286.88	153.13	16997.88	0.00	138.23	2.87

Table 17: Orders registered by phone. Page 65

Name	Type	Instances completed	Instances started	Min. time (s)	Max. time (s)	Avg. time (s)	Total time (s)	Min. time waiting resource (s)	Max. time waiting resource (s)	Avg. time waiting for resource (s)
Home Burguer Lisbon center	Process	45	45	48.58	62.03	55.71	2506.92			
Order received	Start event	45								
Order imported to the systems' database	Task	45	45	18.45	21.85	20.00	900.18	0	0	0
Check costumer's address	Task	45	45	8.20	11.98	9.93	447.03	0	0	0
Is this the appropriate store?	Gateway	45	45							
Process order	Task	37	37	20.76	29.88	25.13	929.89	0	0	0
Check if the order can go to another store	Task	8	8	18.30	20.92	19.76	158.08	0	0	0
Order fulfilled	End event	37								
Can the order go to another store?	Gateway	8	8							
Cancel order and send invite to other store	Task	6	6	8.29	11.85	10.41	62.47	0	0	0
Cancel order	Task	2	2	4.48	4.79	4.63	9.26	0	0	0
Order canceled and invite to other store sent	End event	6								
Order cancelled	End event	2								

Table 18: Orders registered by website. Page 66